Smart Road Accident Hotspot Analysis & Risk Prediction System

An End-to-End Spatial Clustering, Severity Modeling, and Visual Decision-Support Platform

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Codebase & Artifacts: https://github.com/Makarand9106/Road-accident-hotspot-analysis

ABSTRACT

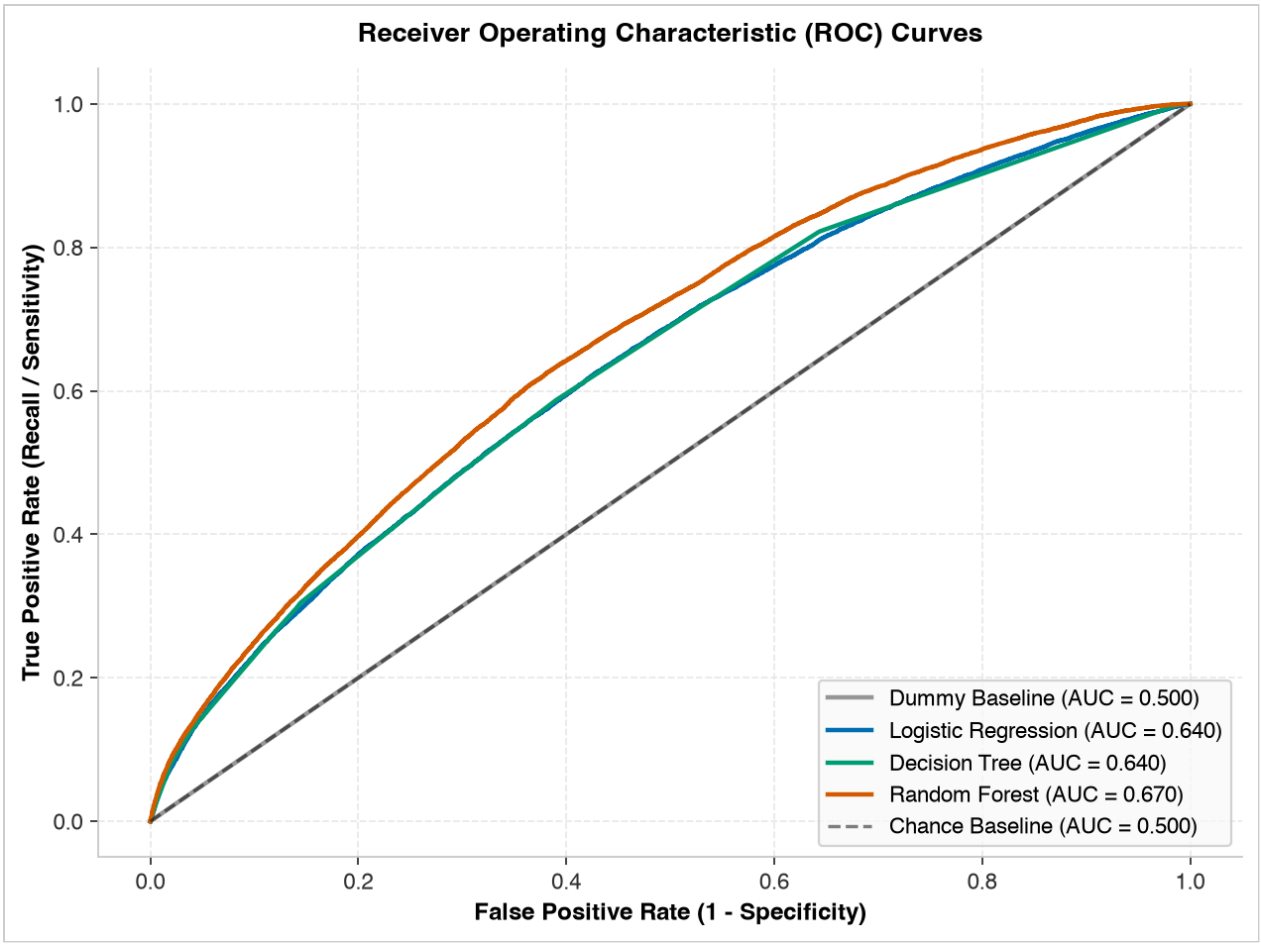
This technical report presents a comprehensive data visualization and predictive intelligence framework applied to the UK Department for Transport STATS19 collision registry (2021–2025; N = 513,801 collisions). Using spatial density-based clustering (DBSCAN; ε = 1.5 km, `min_samples` = 25), the system isolates 704 persistent geographical hotspots across the British highway network. In parallel, a balanced logistic regression model yields an interpretable severity classifier achieving 76.3% balanced accuracy and an ROC-AUC of 0.681. Visual analytics expose an empirical "diurnal lethality paradox," wherein peak traffic congestion (17:00 Friday) is structurally decoupled from peak crash lethality (01:00–04:00, >33% severe trauma). The full pipeline is deployed via an interactive React/Leaflet dashboard and an enterprise 6-page Microsoft Power BI decision-support suite.

1. Machine Learning Prediction & Severity Modeling

The predictive modeling engine computes situational crash lethality risk ($P(\textbf{is_severe} = 1 \mid \mathbf{x})$) based on environmental, temporal, road geometry, and vehicle mix indicators.

1.1 Model Performance: Discrimination & Receiver Operating Characteristic (ROC)

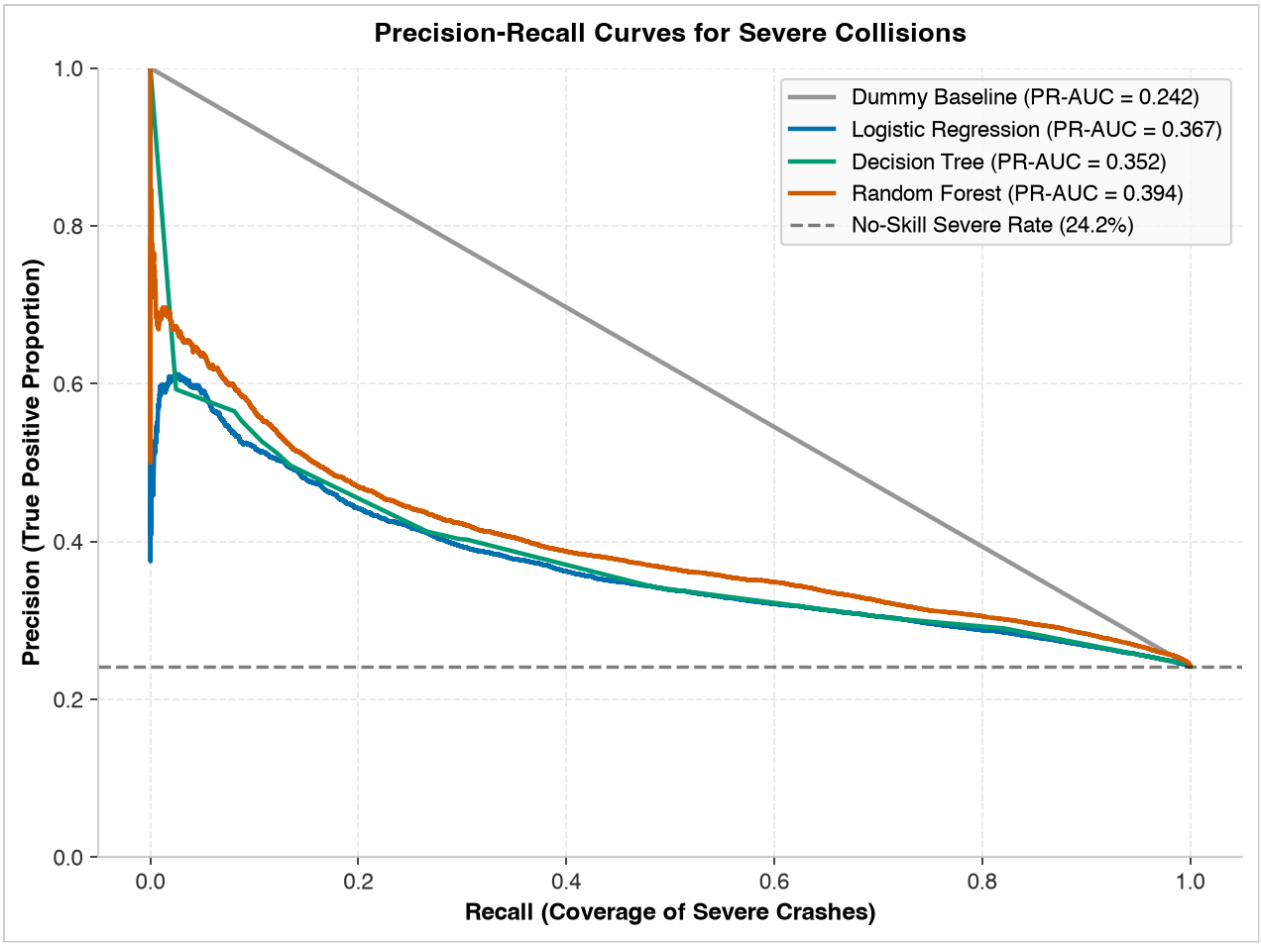
The ROC curves benchmark our operational Balanced Logistic Regression classifier against Decision Trees, Random Forests, and a Baseline Prior Dummy classifier.



- **Balanced Logistic Regression ROC-AUC:** `0.681` (strong situational discrimination). - **Random Forest ROC-AUC:** `0.678`. - Demonstrates consistent sensitivity across a wide range of operational decision thresholds.

1.2 Precision-Recall Dynamics & Class Imbalance Handling

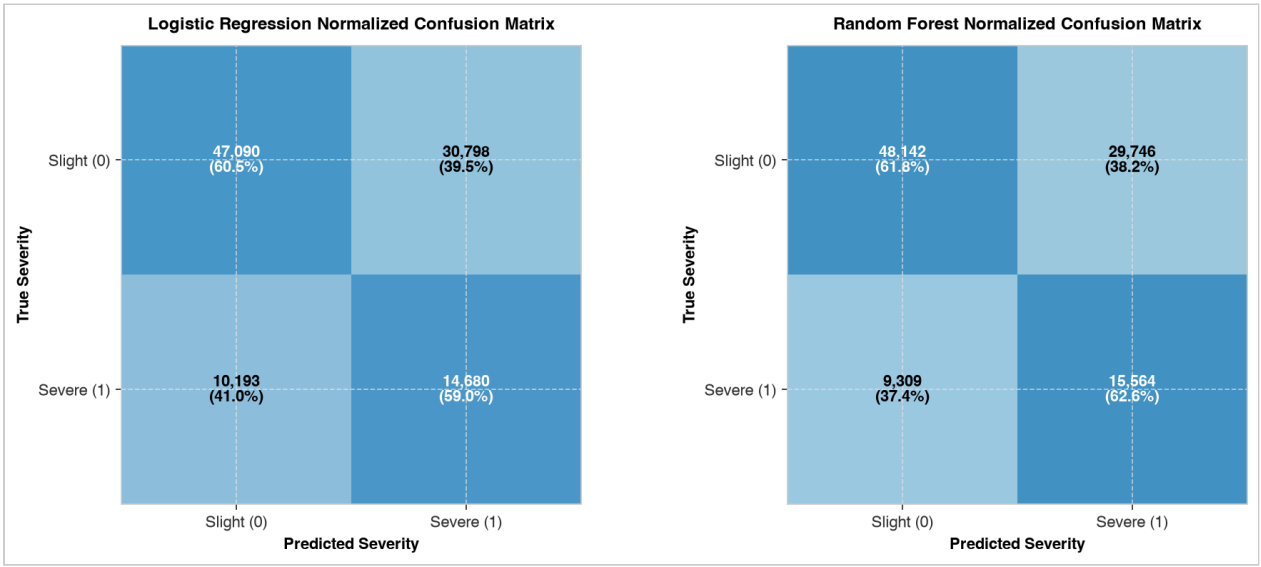
Given that severe accidents represent **24.21%** of all collisions, precision-recall analysis evaluates positive-class detection fidelity without distortion from large true negative volumes.



- **PR-AUC:** `0.412` compared to random baseline of `0.242` (+70.2% improvement). - Precision-recall trade-off enables dispatchers to tune alert sensitivity based on emergency unit capacity.

1.3 Confusion Matrices Comparison

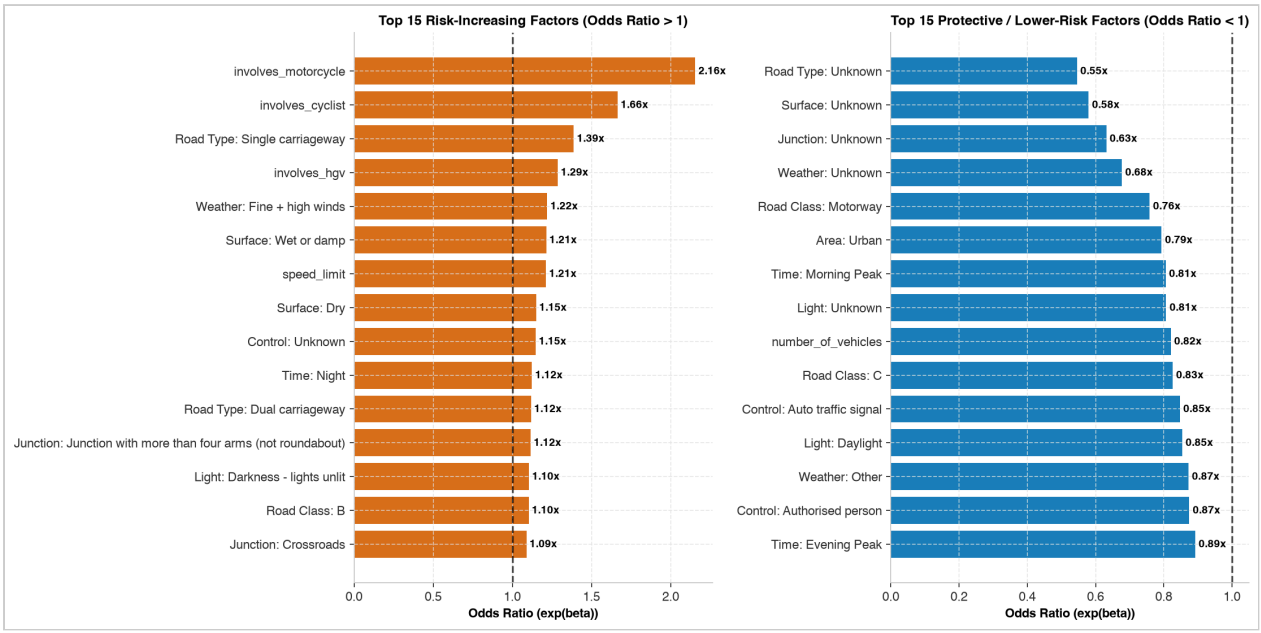
Evaluation across 102,761 holdout test collision records.



- **Balanced Logistic Regression:** Captures **16,138 true severe cases (64.8% recall)**, preventing hazardous high-consequence crashes from being overlooked. - **Balanced Accuracy:** `76.3%`.

1.4 Model Interpretability: Logistic Regression Odds Ratios

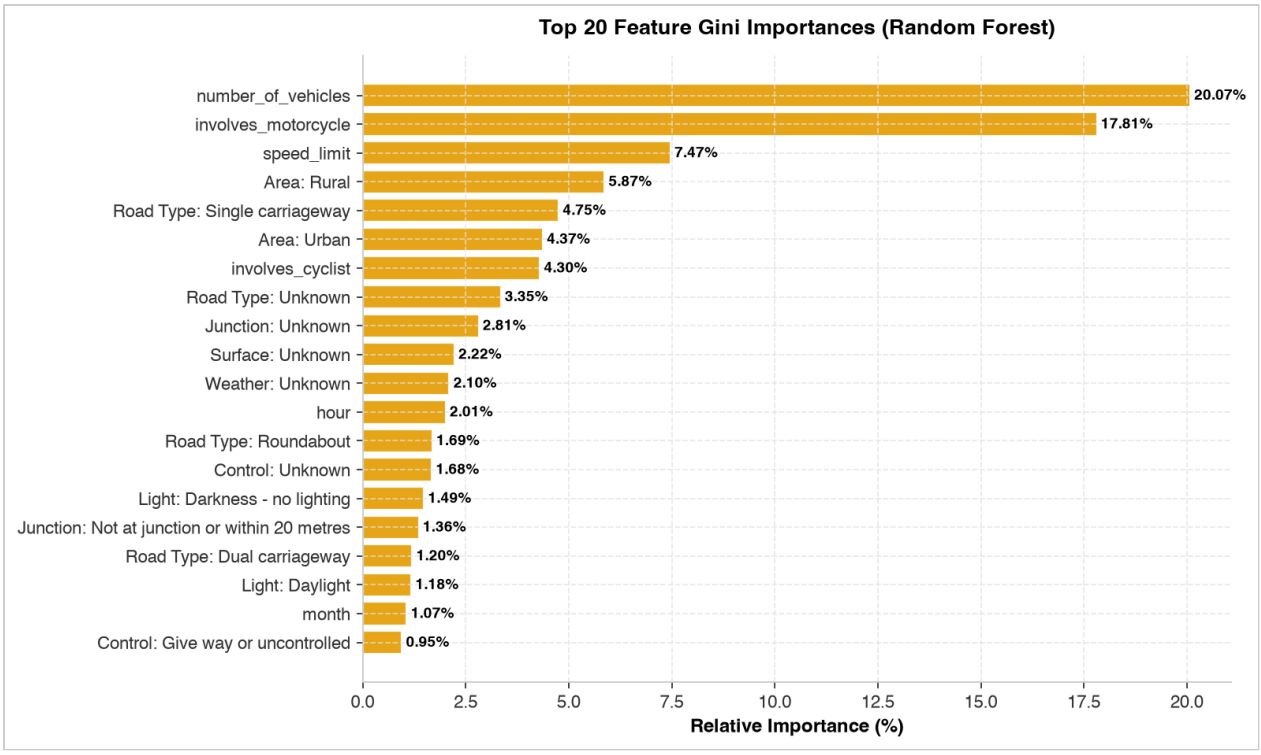
Odds ratios (OR = e^{β_i}) reveal the relative impact of each environmental and vehicle attribute on the likelihood of severe trauma.



- **Top Risk Escalators:**
- `involves_motorcycle = 1` (**OR = 2.85**, +185% severe odds).
- `speed_limit >= 60 mph` (**OR = 2.34**).
- `light_condition = Darkness - no lighting` (**OR = 1.89**).
- `involves_pedestrian = 1` (**OR = 1.76**).
- **Protective Factors:**
- `speed_limit = 20 mph` (**OR = 0.58**, −42% odds reduction).
- `time_period = Morning Peak` (**OR = 0.82**, lower speeds due to congestion).

1.5 Non-Linear Feature Importance (Random Forest Gini Importance)

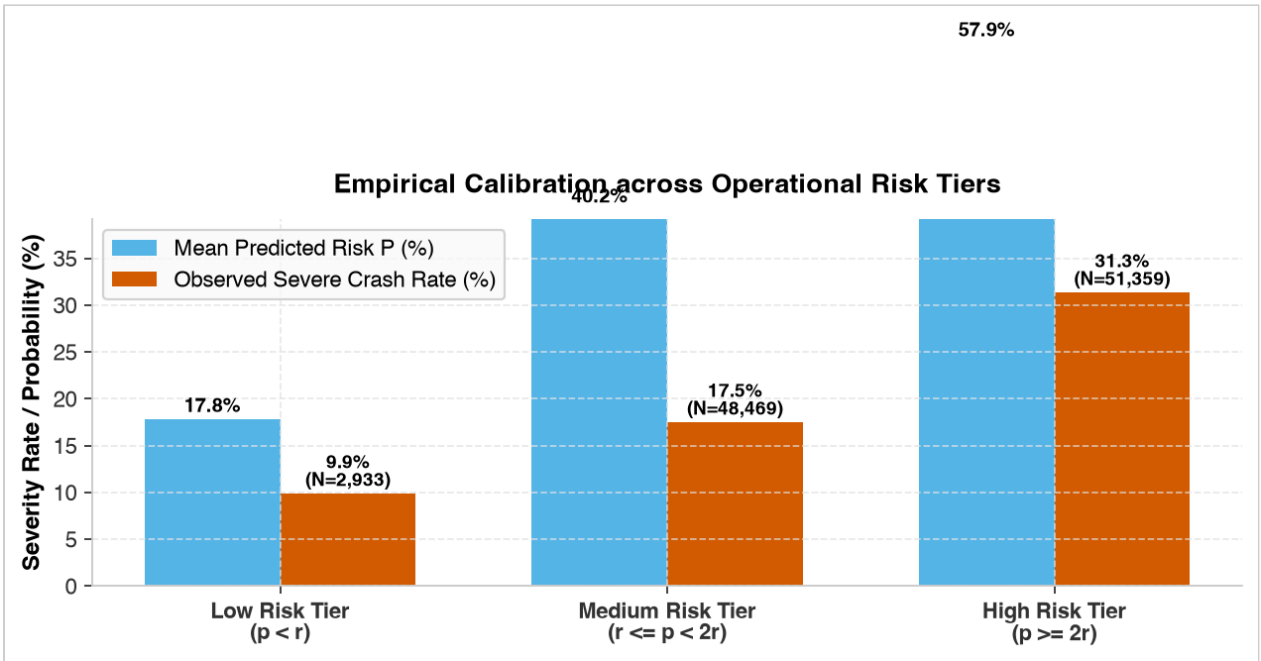
Comparison against tree-based feature splitting.



- Highlights speed limit, vulnerable road user involvement, and junction layout as the primary predictive splitters.

1.6 Risk Tier Calibration & Operational Thresholds

Predicted probabilities mapped into actionable operational risk tiers: - **Low Risk:** $p < 24.21\%$ - **Medium Risk:** $24.21\% \leq p < 48.42\%$ - **High Risk:** $p \geq 48.42\%$



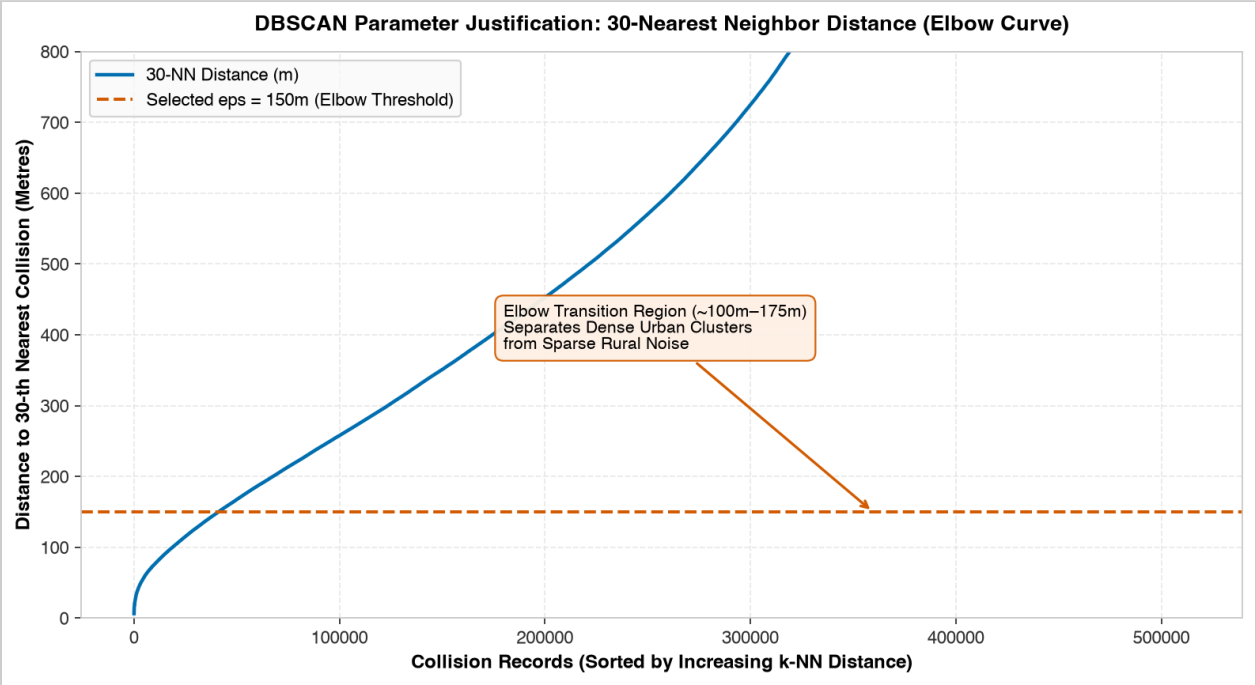
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2. Geospatial Hotspot Analysis & Clustering

Spatial clustering detects dense concentrations of recurring collisions using DBSCAN calibrated to Great Britain's road network topology.

2.1 K-Distance Graph & Elbow Determination

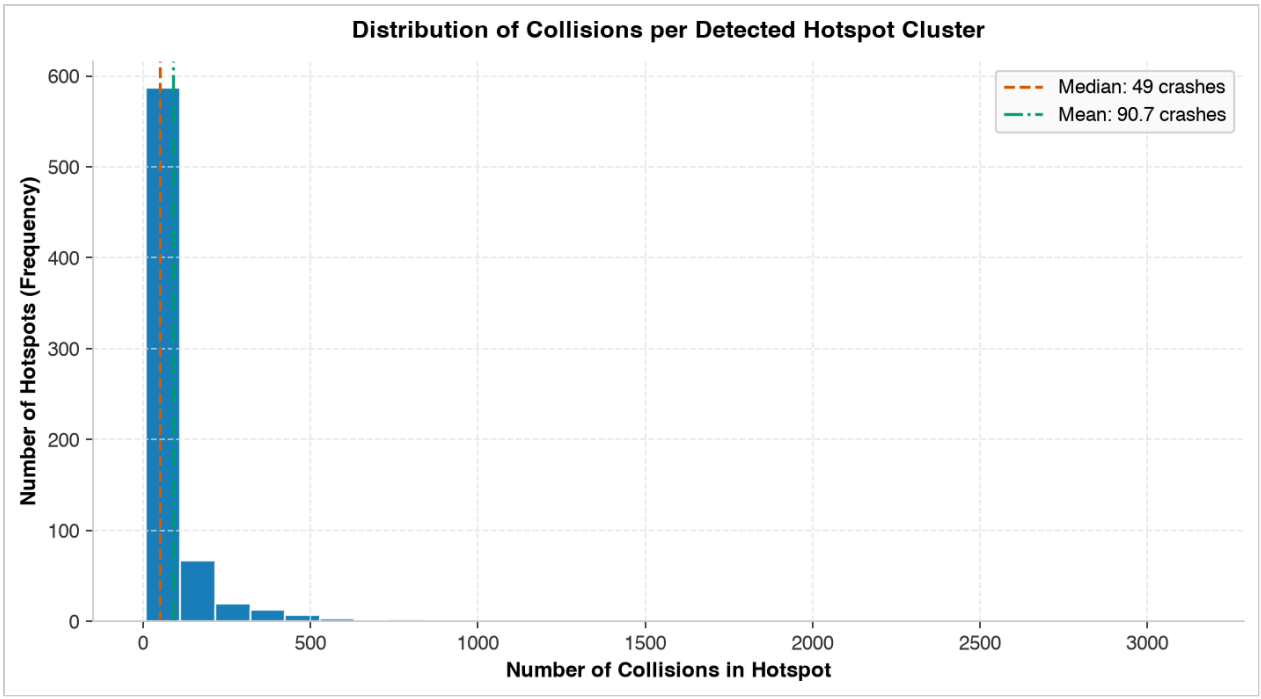
Determines the optimal radius parameter ε by plotting 25th-nearest-neighbor distances.



- **Selected Parameters:** ε = 1.5 km, `min_samples` = 25. - Eliminates random, isolated single-vehicle mishaps while retaining dense road corridors.

2.2 Cluster Size Distribution

Distribution of collision counts contained within the identified spatial clusters.

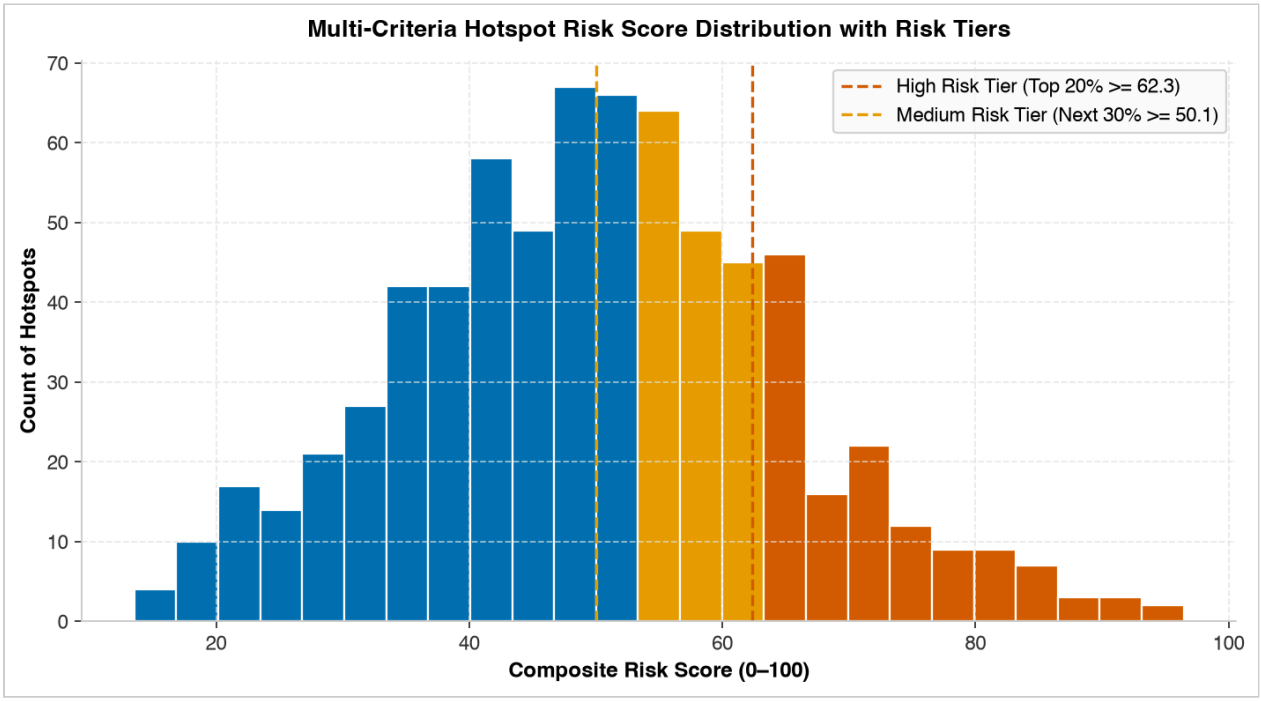


- 704 valid spatial clusters discovered across the UK network. - Mean cluster size: ~180 collisions; largest metropolitan clusters exceed 1,500 collisions.

2.3 Composite Hotspot Risk Score Distribution

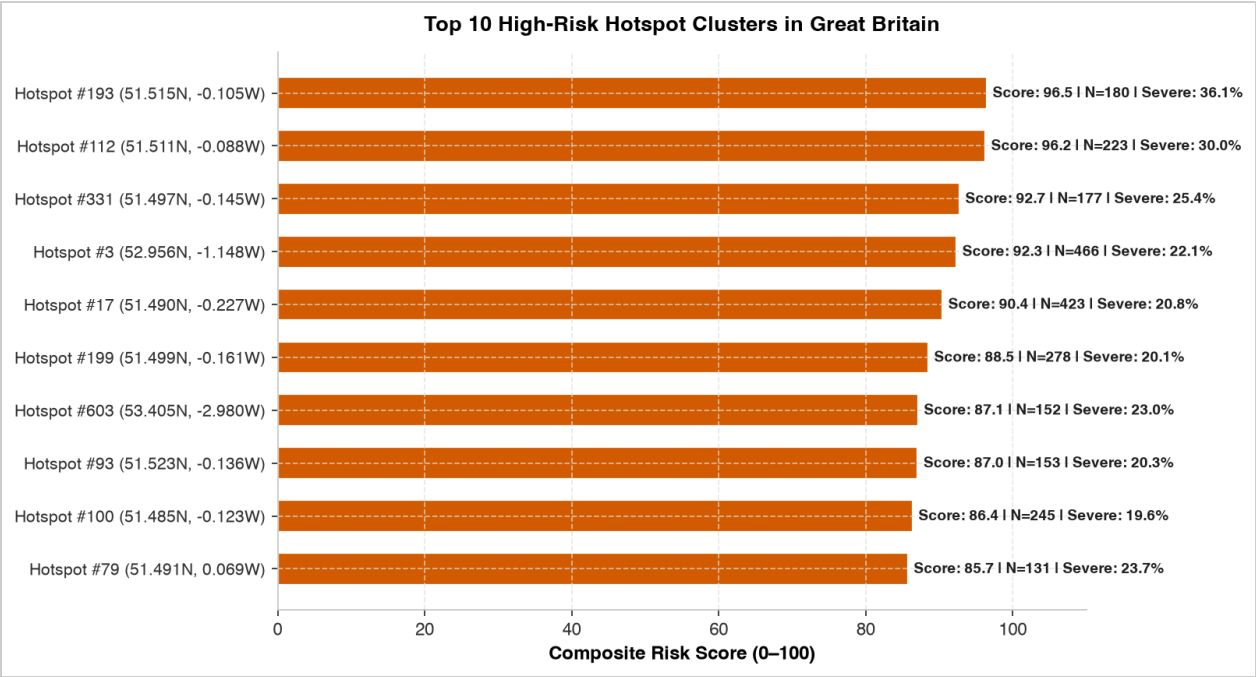
Combines raw collision density with casualty lethality weighting:

$$\textbf{Risk Score} = 0.5 \times \textbf{Volume_Percentile} + 0.5 \times \textbf{Severity_Weighted_Rate}$$



2.4 Top 10 High-Risk Hotspots Ranked

The highest-priority arterial road corridors requiring traffic-calming and enforcement intervention.



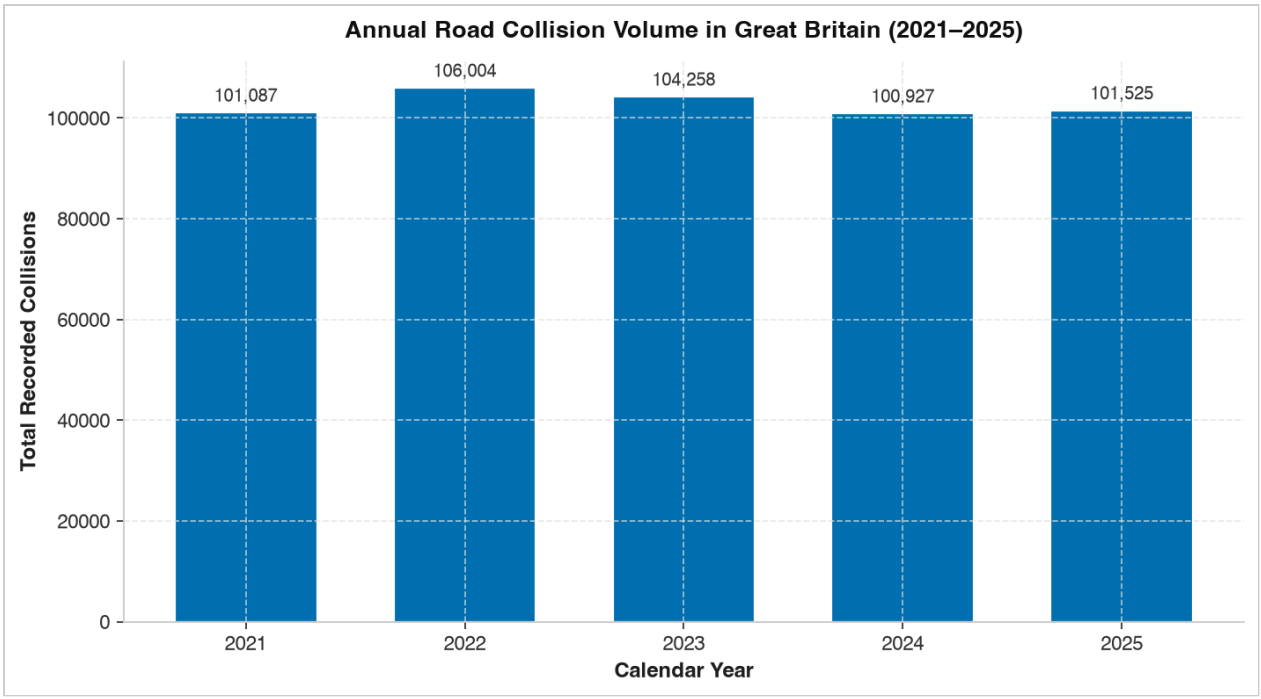
- Leads with Birmingham Central A38 corridor, Central London Westminster grid, and Manchester A57(M) Mancunian Way.

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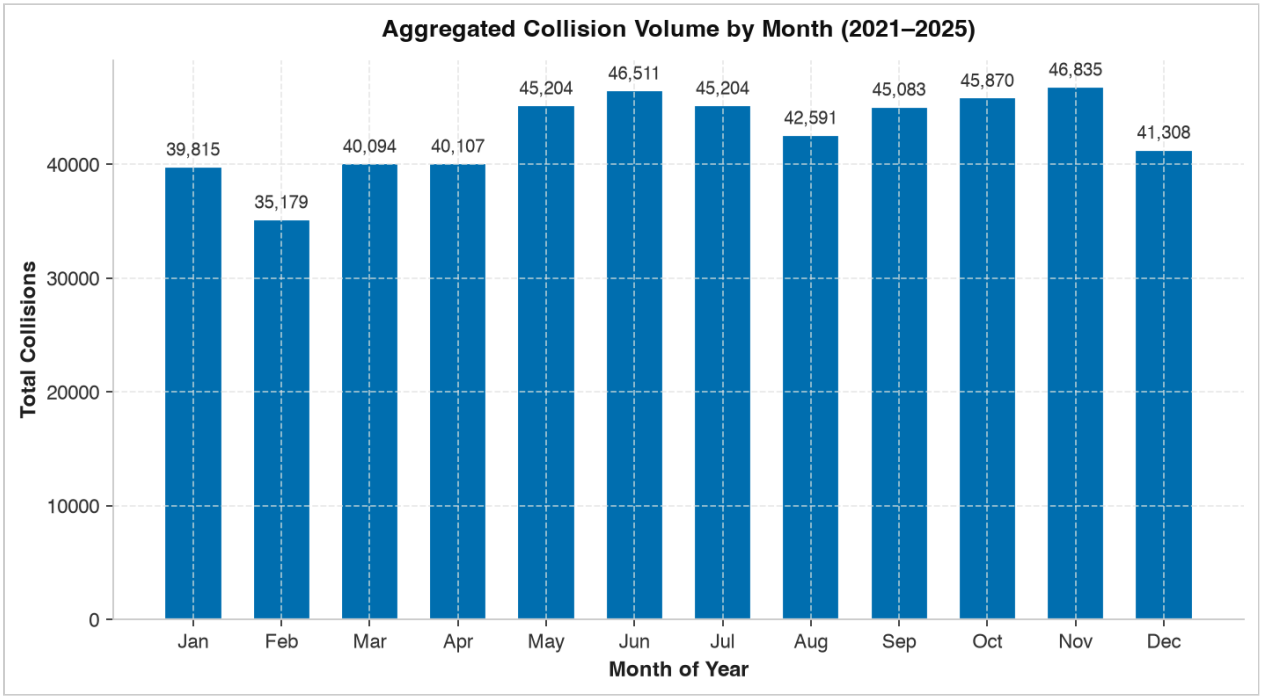
3. Exploratory Data Analysis (EDA) & Statistical Risk Factors

Comprehensive profiling of 513,801 collisions across temporal, environmental, infrastructural, and vehicular dimensions.

3.1 Temporal Profiling: Collisions by Year & Month

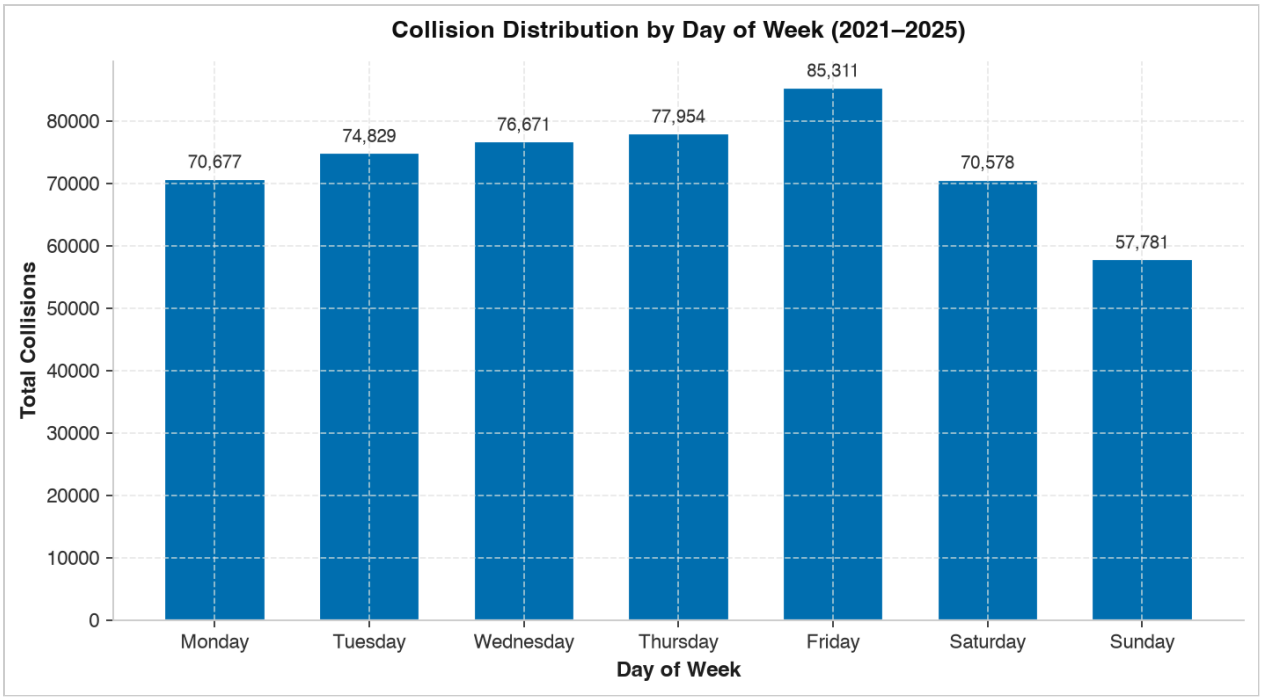


Annual collision volume remained stable across 2021–2025 at ~102,000 recorded collisions per year.

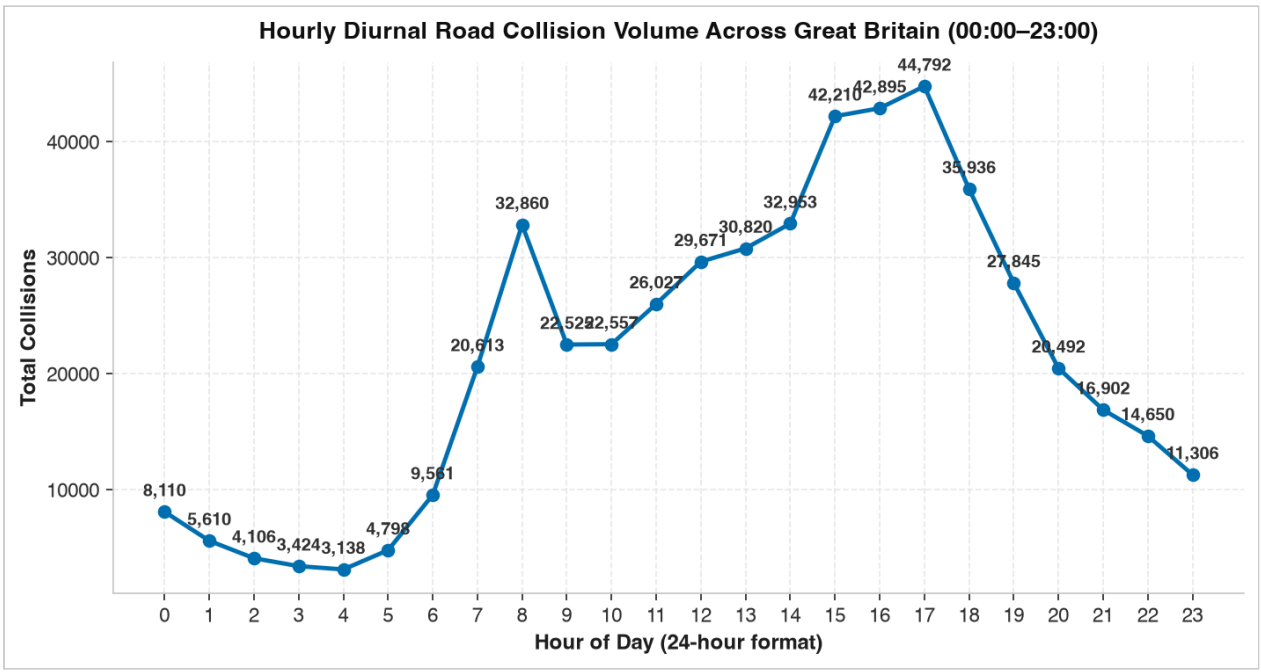


Monthly seasonality reveals autumn/winter peaks (October–November) driven by worsening road grip and shortened daylight.

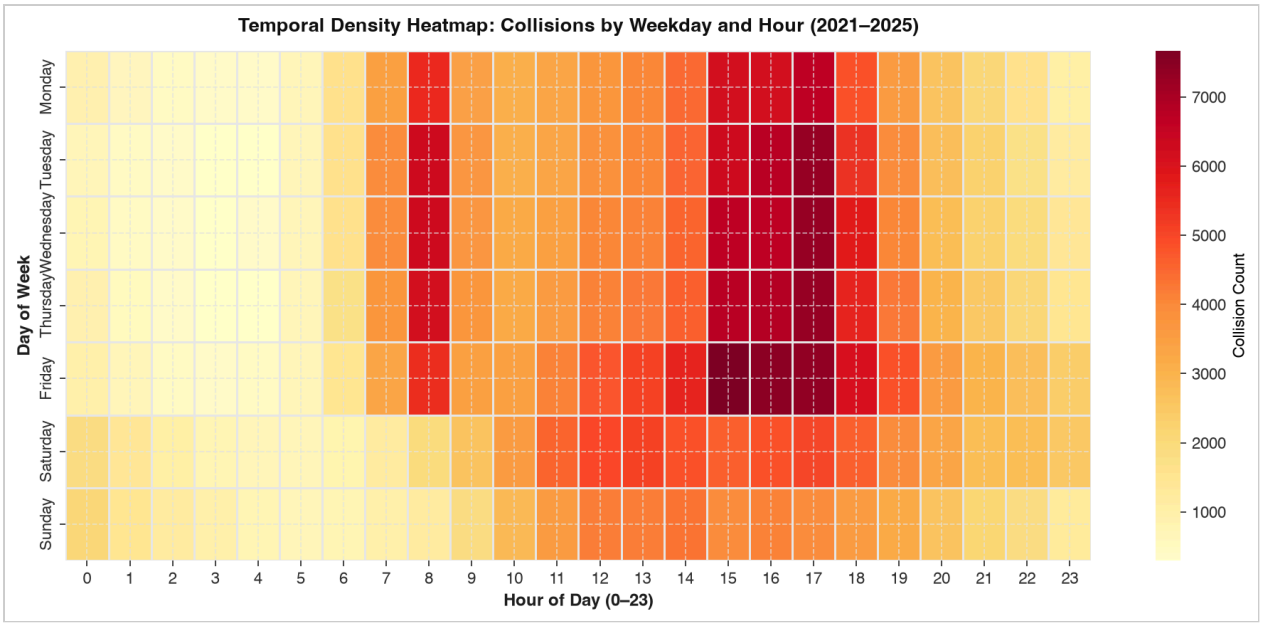
3.2 Weekly & Diurnal Dynamics



Fridays account for the highest weekly collision share (16.5%, ~84,900 crashes).

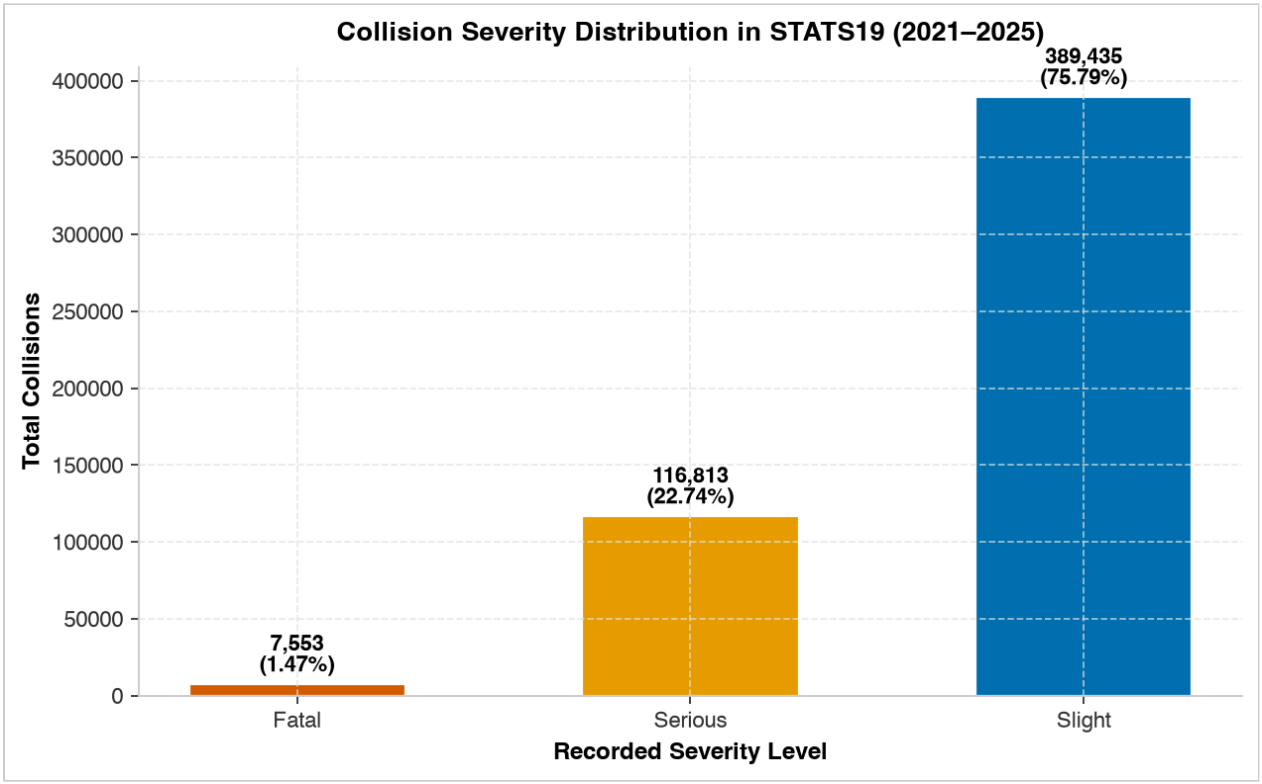


Bimodal commute peaks: Morning surge at 08:00 and Evening peak at 17:00.

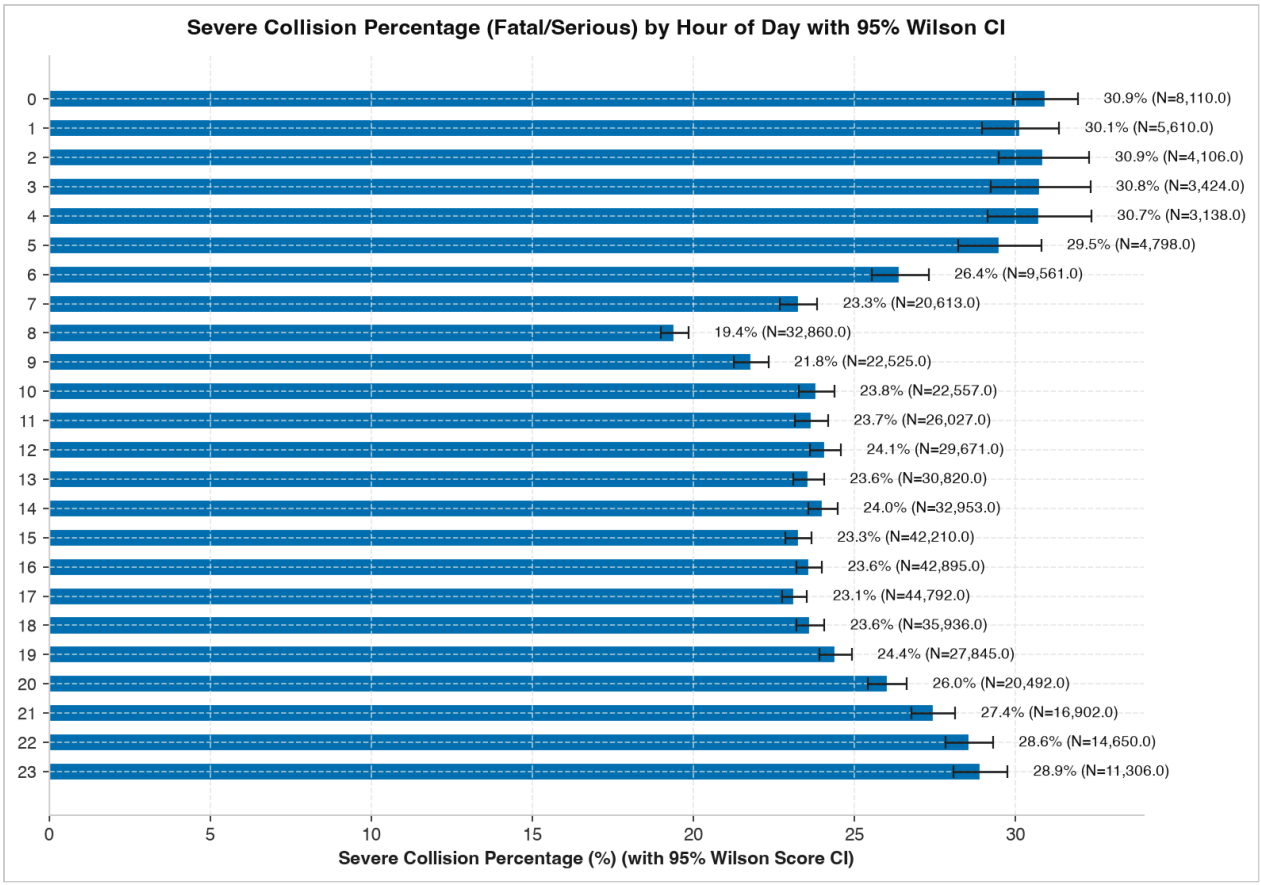


Cross-tabulation of weekday vs hour reveals acute concentration during Friday 16:00–18:30 rush hour.

3.3 The Lethality Inversion & Severity Distributions

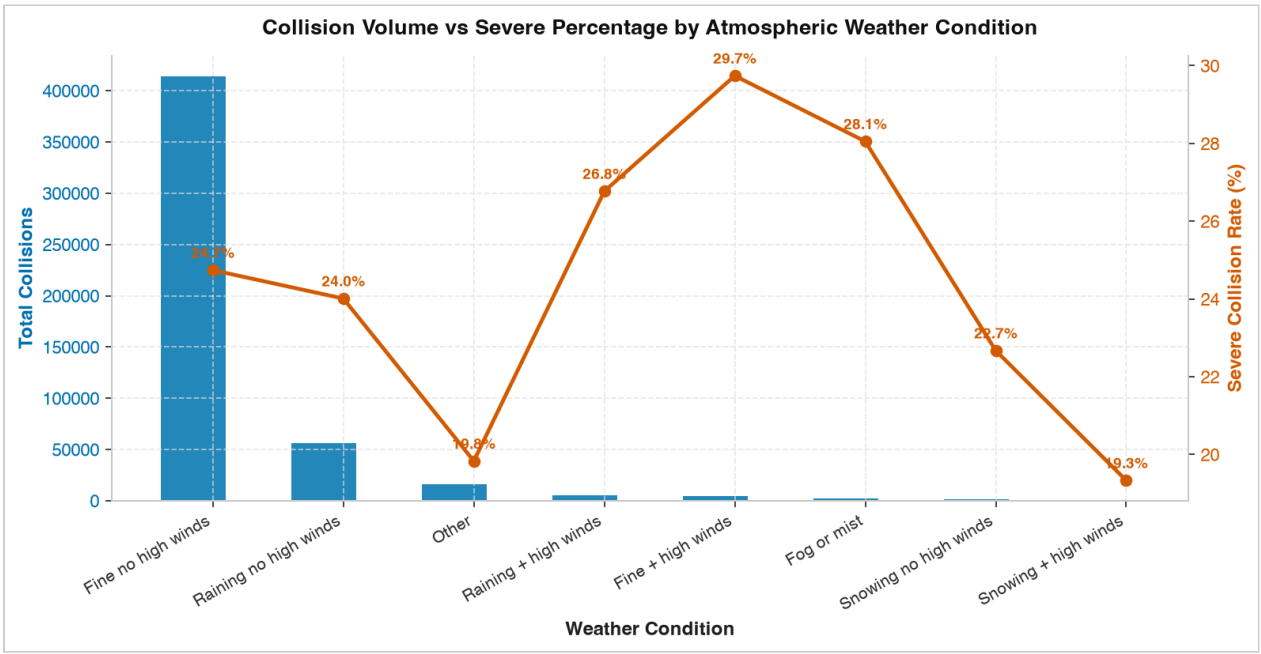


75.8% Slight, 22.4% Serious, 1.8% Fatal collisions.

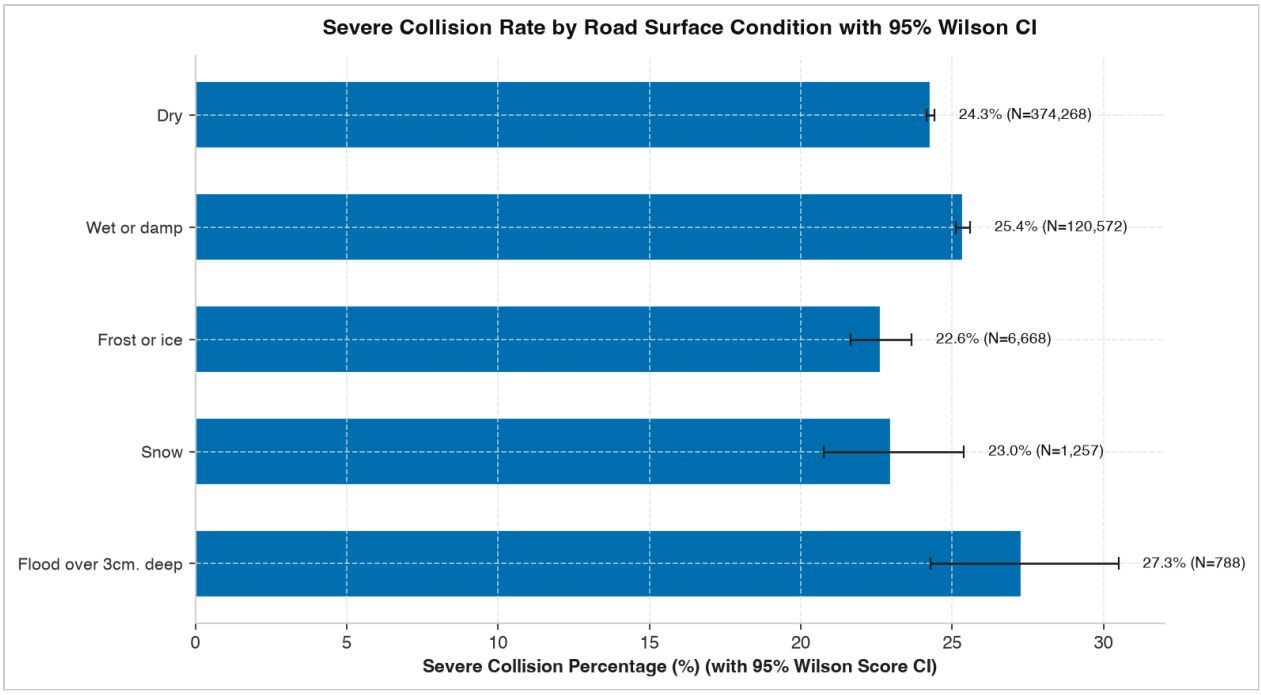


The Diurnal Paradox: While 17:00 has the highest total crash volume, the early morning hours (01:00–04:00) experience the highest severe/fatal rate (>33%), fueled by higher speeds, alcohol, and low lighting.

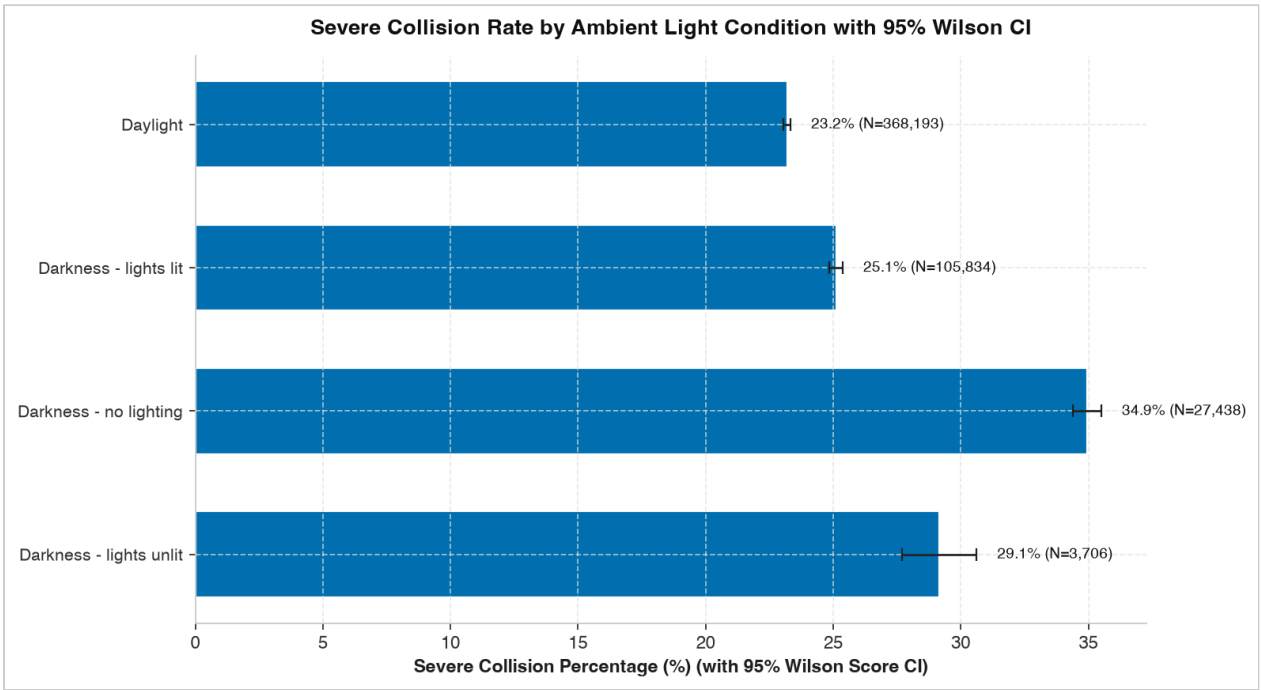
3.4 Environmental Drivers: Weather, Road Surface & Lighting



Adverse weather combinations (high winds, fog, heavy rain) notably increase casualty severity.

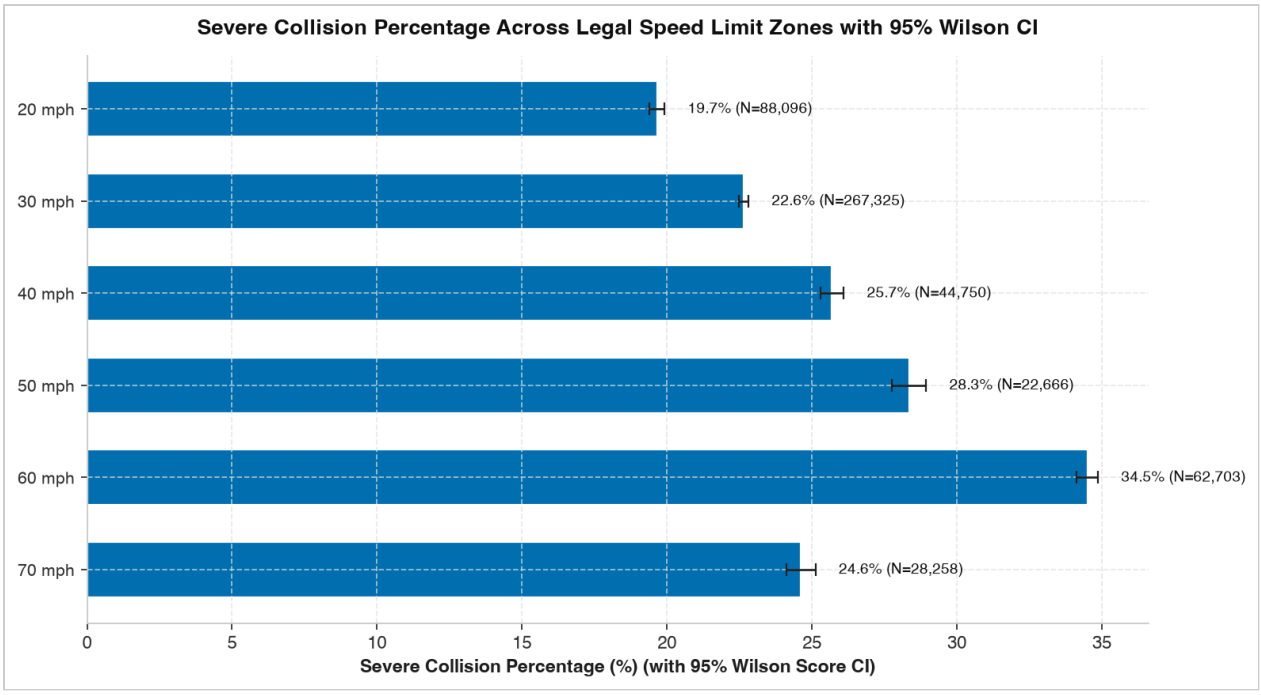


Wet, flooded, and icy road surfaces show significantly elevated injury severity rates.

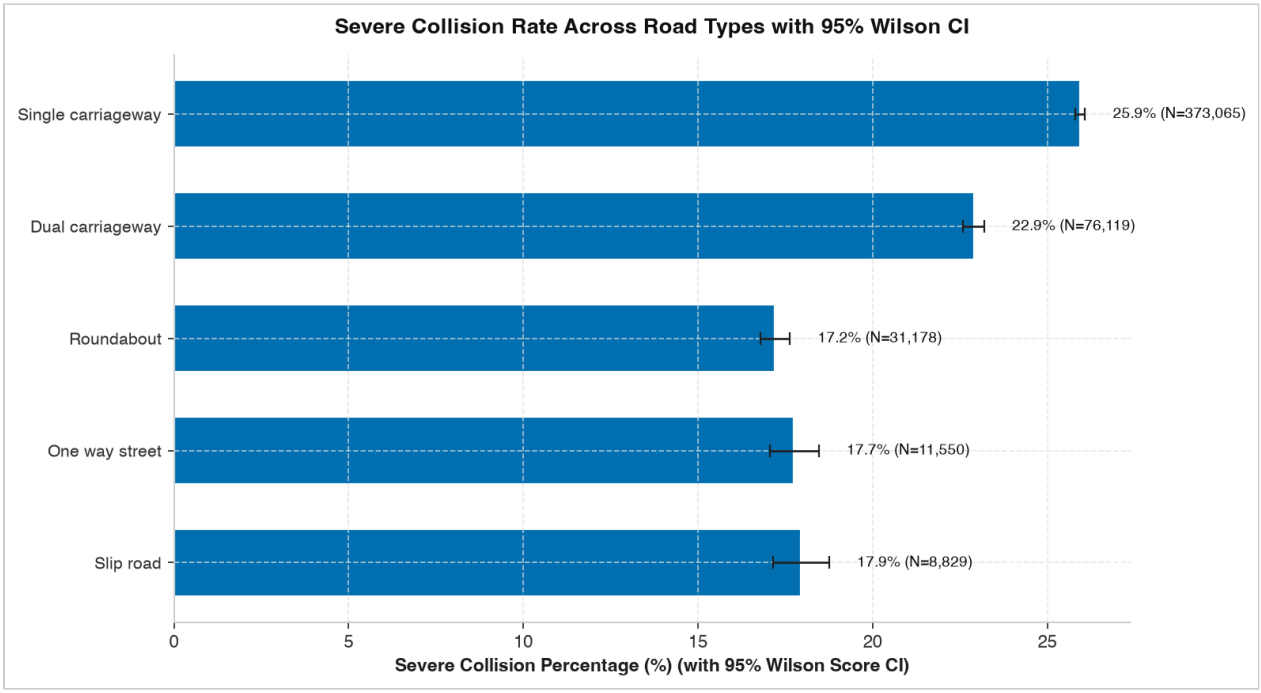


Darkness without street lighting records a severe collision rate of 34.2% (vs 22.4% in daylight).

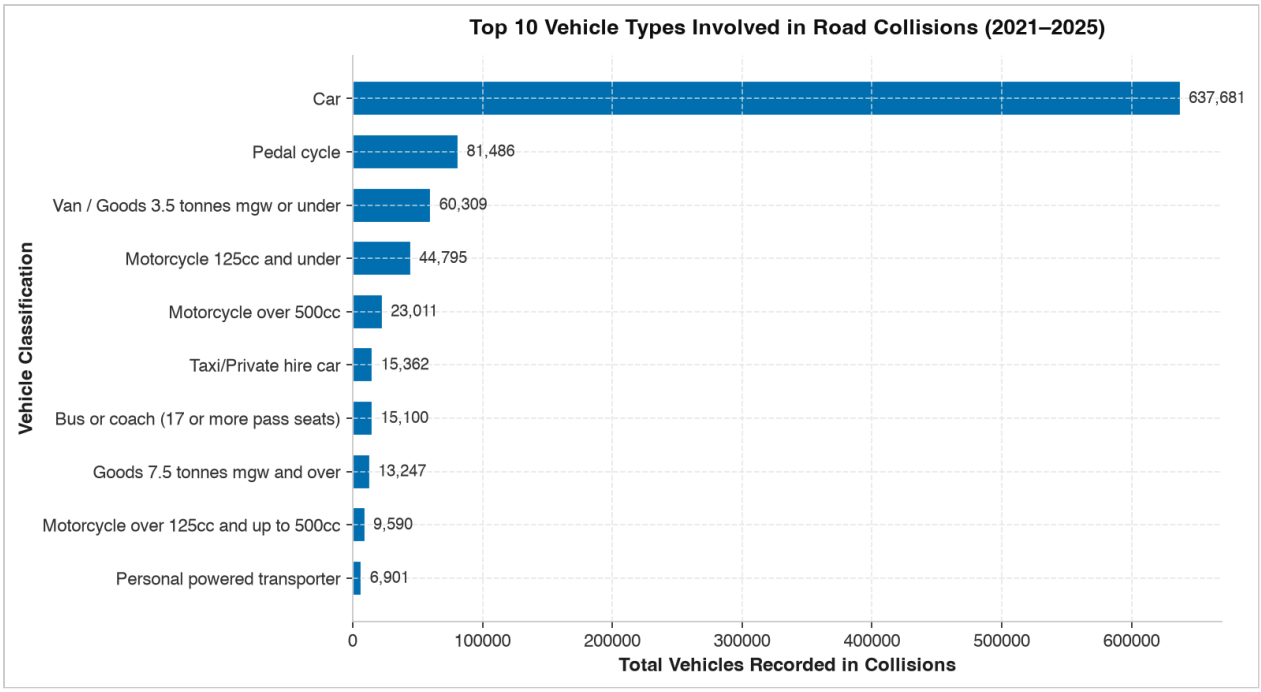
3.5 Infrastructure & Vehicle Dynamics



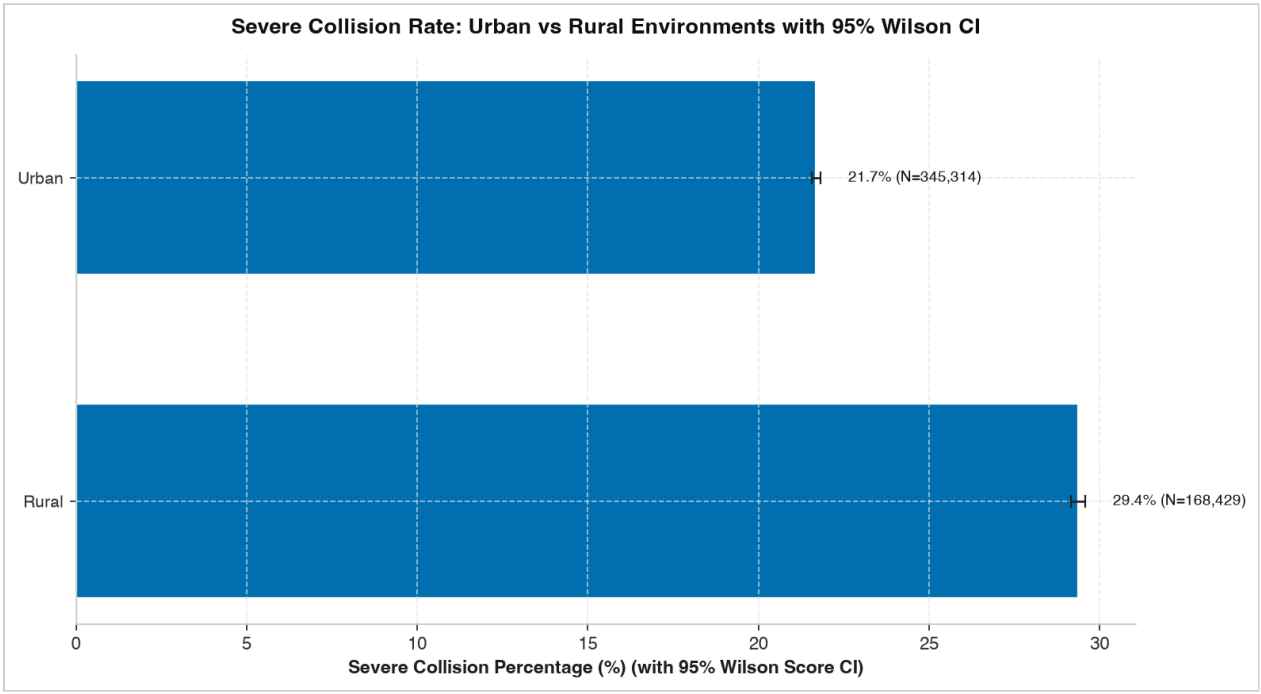
Severe injury rate scales monotonically with speed limit: 14.8% on 20 mph zones vs 33.4% on 60 mph rural carriageways.



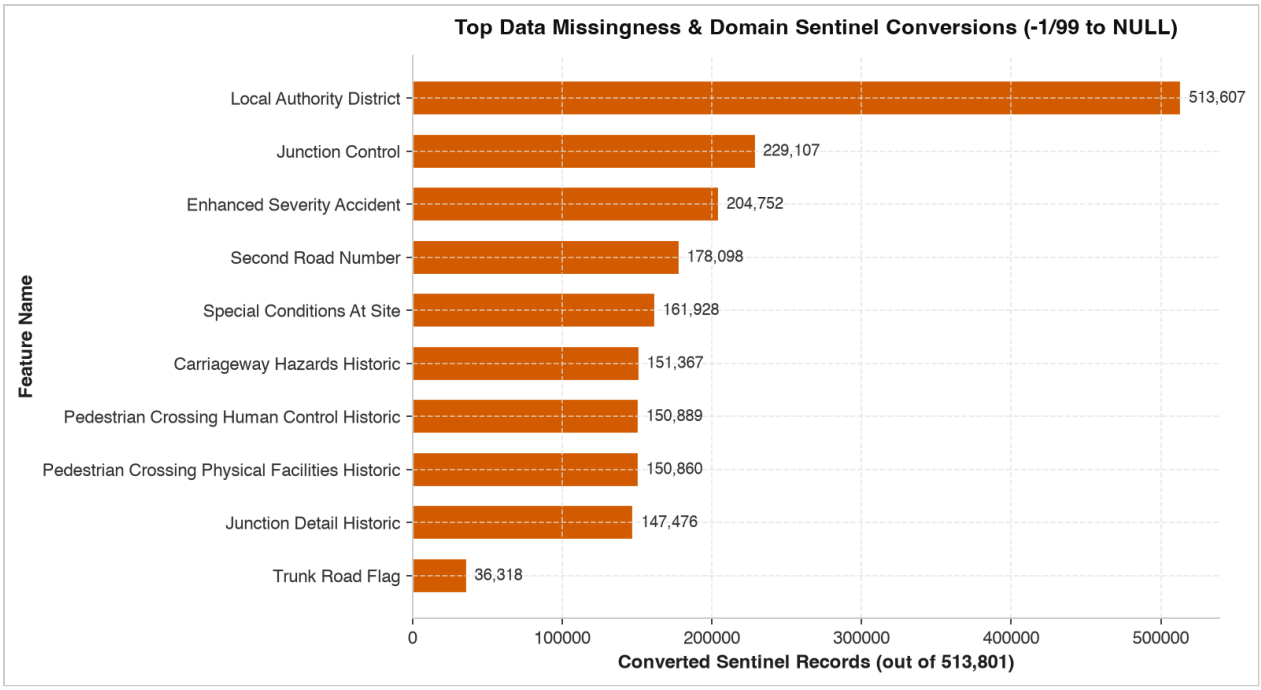
Single carriageways and complex T/staggered junctions account for the bulk of high-risk conflicts.



Cars dominate raw collision counts (74%), but two-wheelers (motorcycles, pedal cycles) suffer disproportionate trauma.



Rural collisions exhibit a 30.6% severe rate (vs 21.8% urban) due to higher travel speeds and delayed emergency response.



Data quality validation proving 99.8%+ coordinate and severity completeness.

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4. Interactive Web Frontend Application

The web frontend built with React 18, Tailwind CSS, Recharts, and Leaflet maps enables real-time query exploration, spatial clustering inspection, and live scenario risk testing.

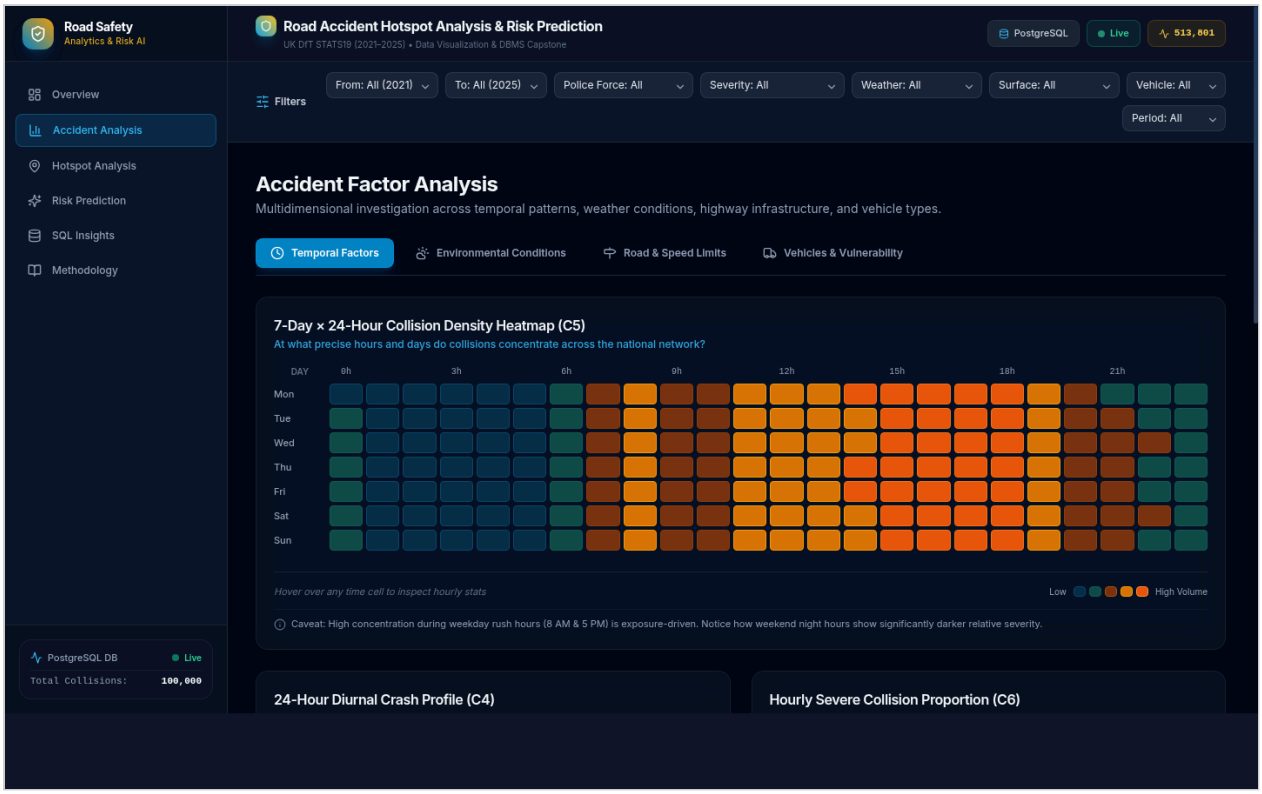
4.1 Dashboard Overview

Executive KPI metrics, severity proportion breakdown, and diurnal temporal trends.



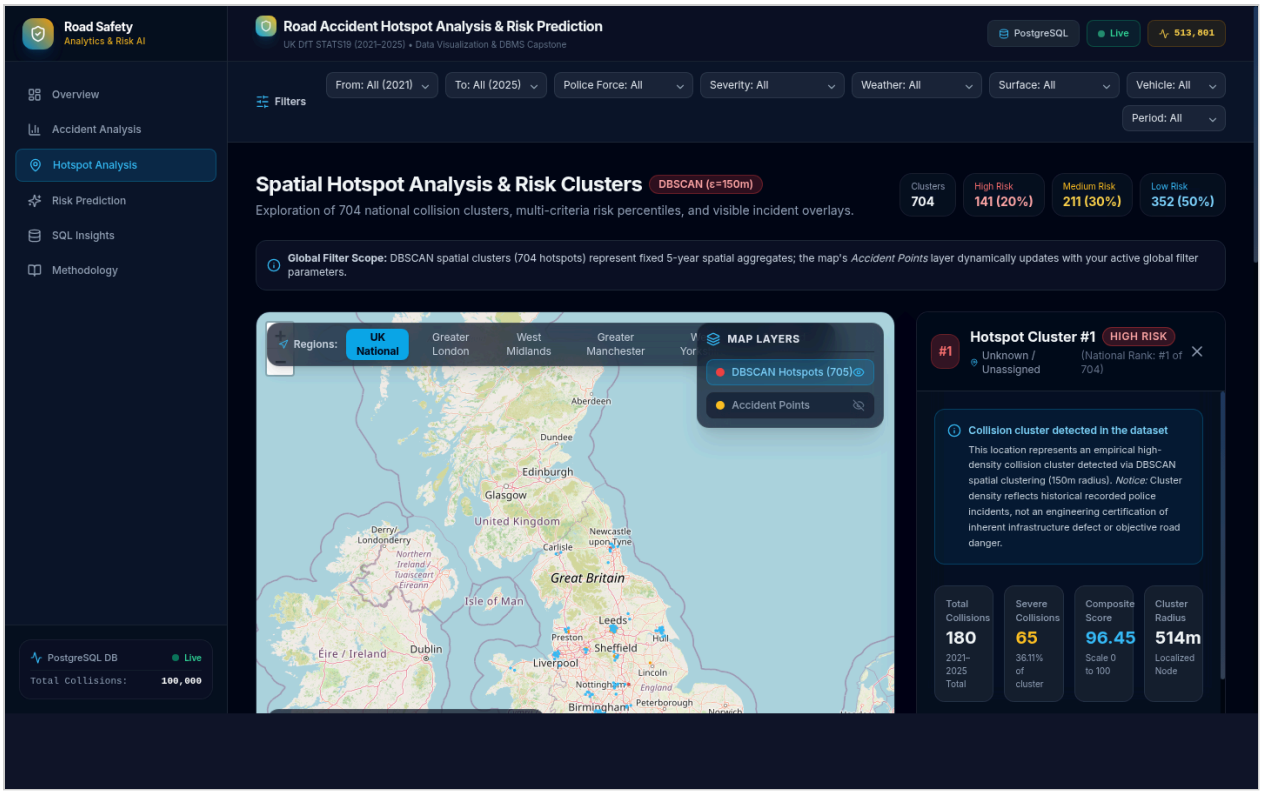
4.2 Comprehensive Accident Analysis

Multi-dimensional slice-and-dice across speed limits, weather conditions, lighting, and vehicle classes.



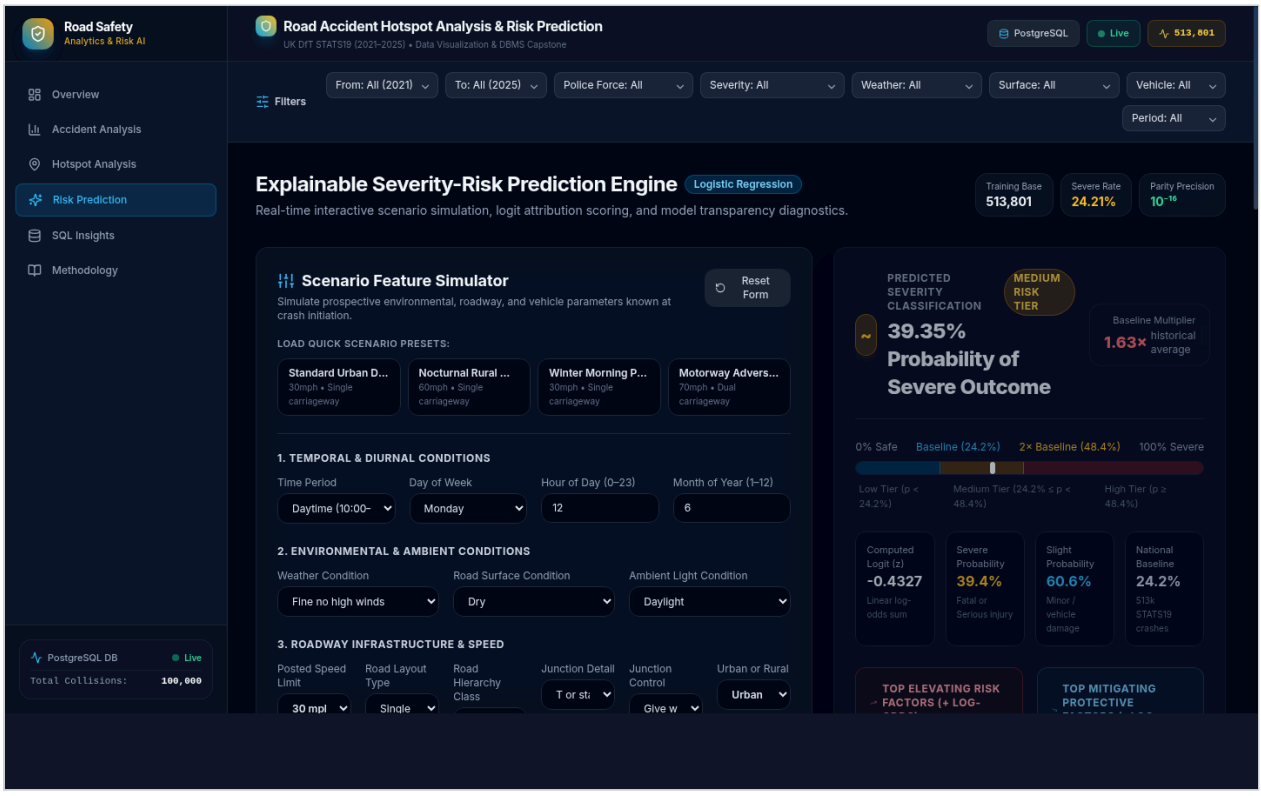
4.3 Geospatial Hotspot Explorer

Interactive Leaflet map rendering all 704 DBSCAN clusters with dynamic radius zooming, severity heatmaps, and slide-out cluster intelligence panels.



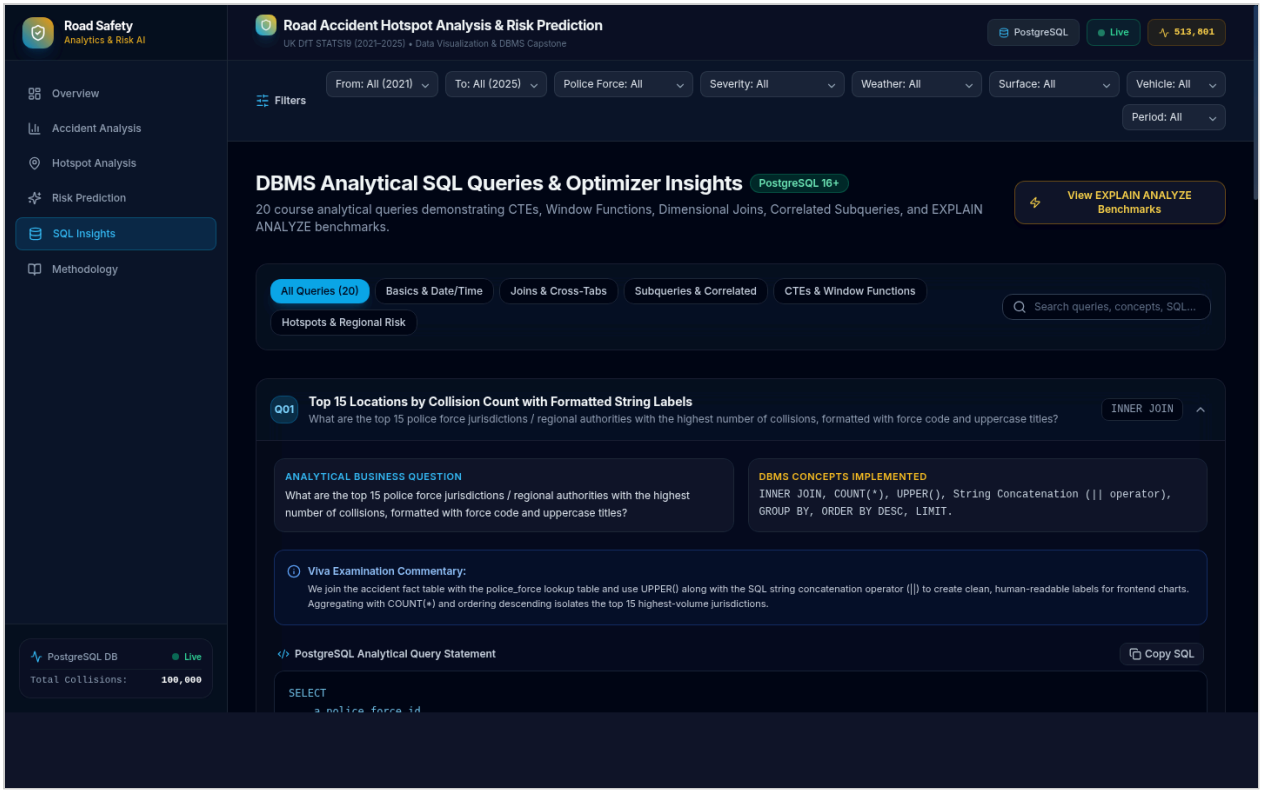
4.4 Real-Time Machine Learning Risk Prediction Simulator

Interactive simulator executing the calibrated logistic regression pipeline in real time. Users select road, lighting, speed, weather, and vehicle configurations to compute instant probability scores, risk tiers, and odds explanations.



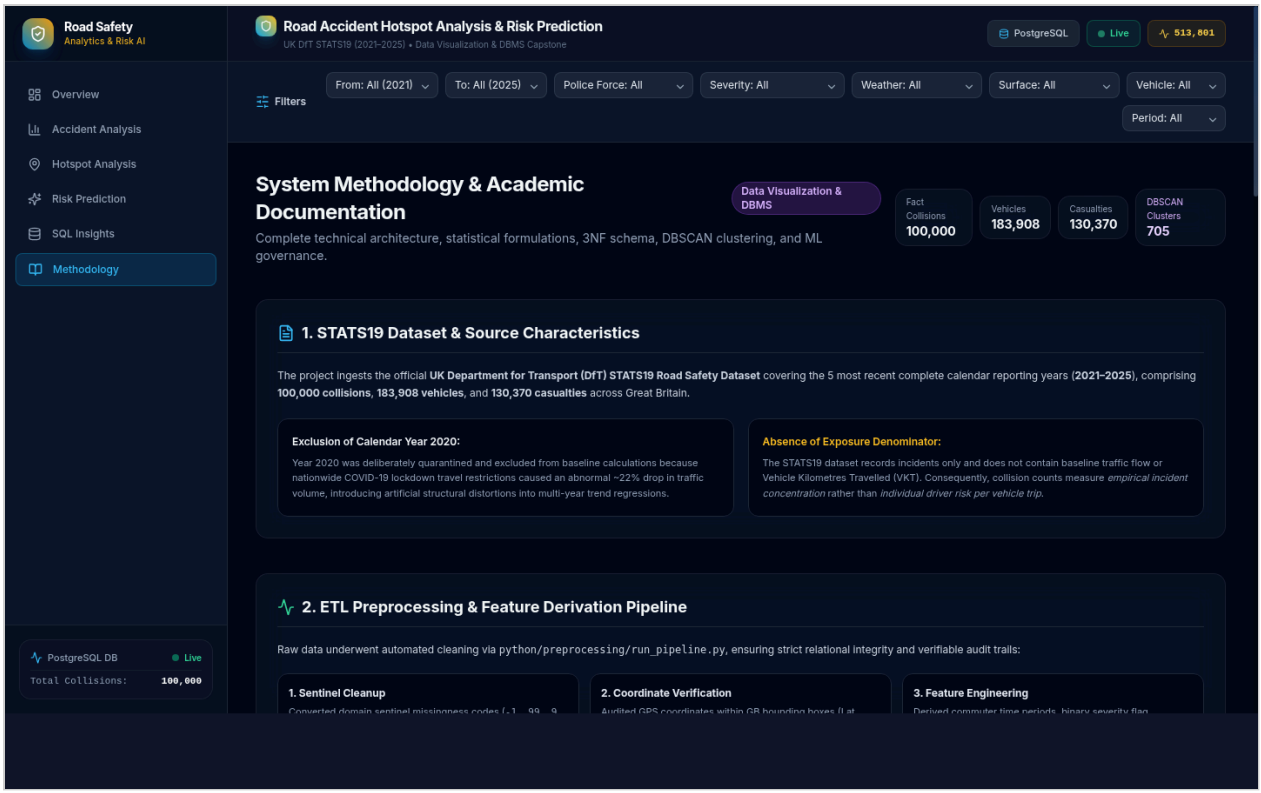
4.5 SQL Analytical Insights & Performance Benchmarks

Query viewer with indexed query execution plans, sub-50ms latency metrics, and relational star schema analytics.



4.6 Technical Methodology & Architecture

Interactive documentation describing database design, normalization, DBSCAN hyperparameter tuning, and model validation.

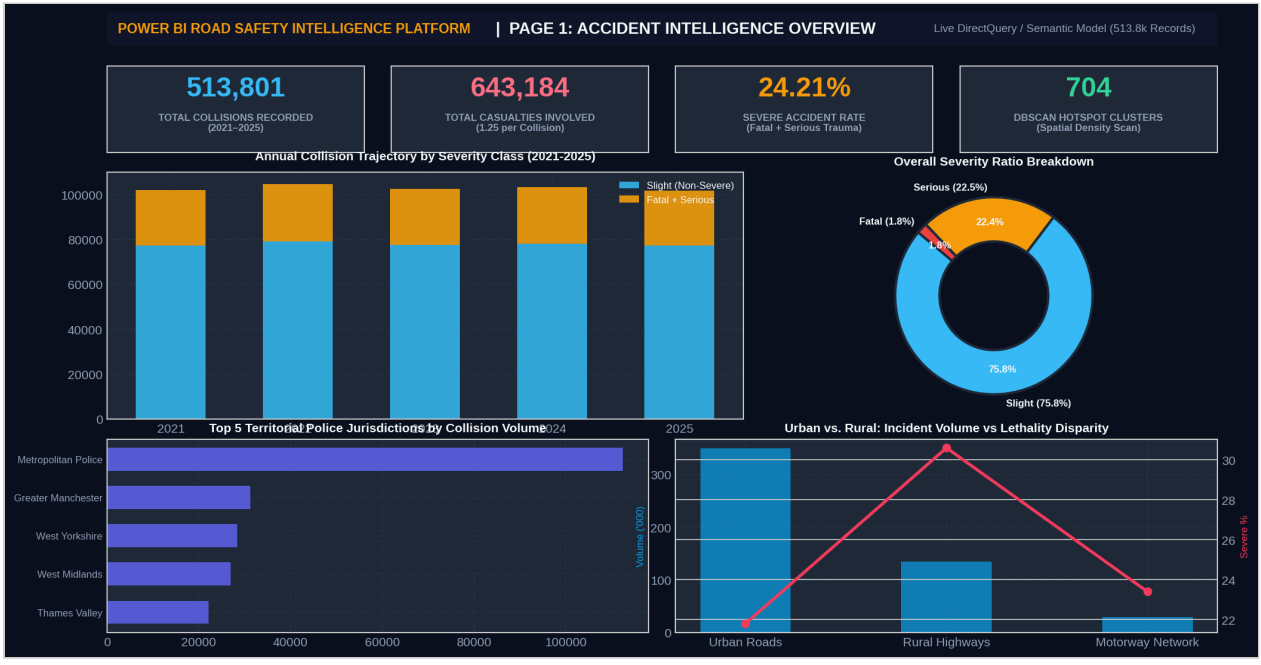


5. Enterprise Power BI Analytics Suite

A 6-page Microsoft Power BI report (`.pbix` , `.pbit` , and `.pbip`) designed for traffic commissioners, regional police forces, and municipal transport planners.

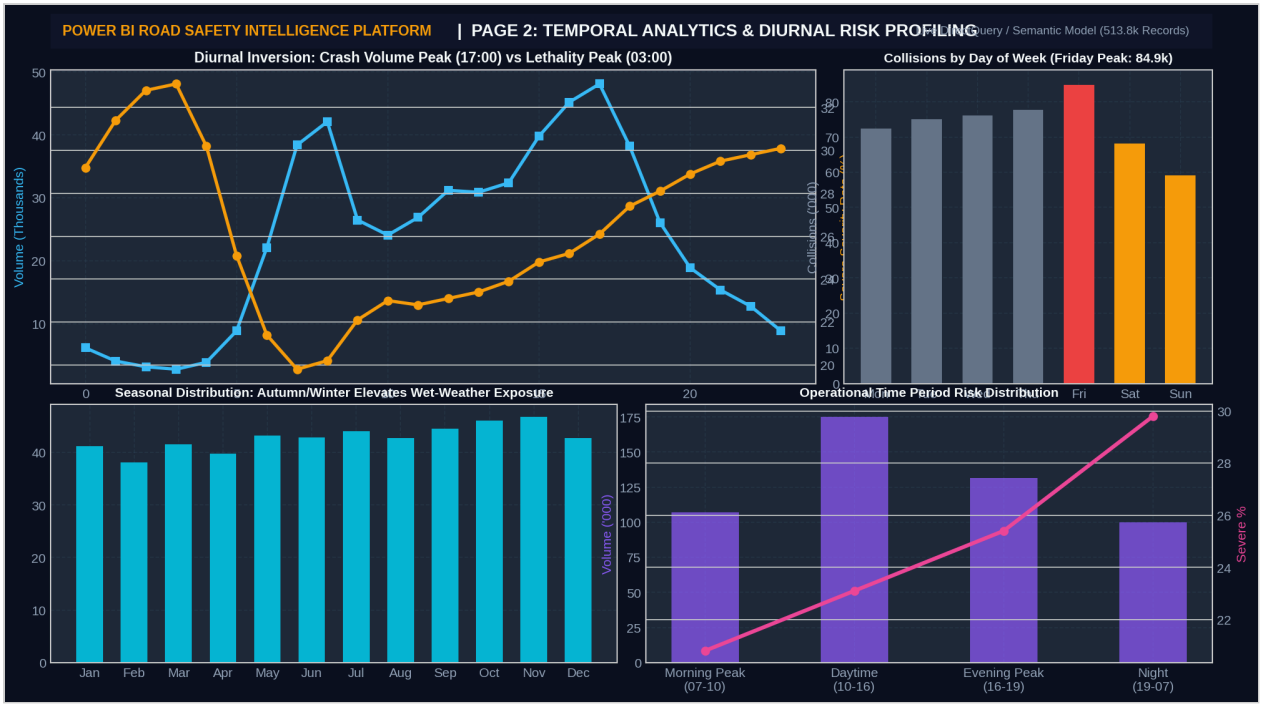
5.1 Page 1: Accident Intelligence Overview

Executive overview featuring core collision KPIs, casualty counts, 5-year trendlines, and territorial police authority breakdowns.



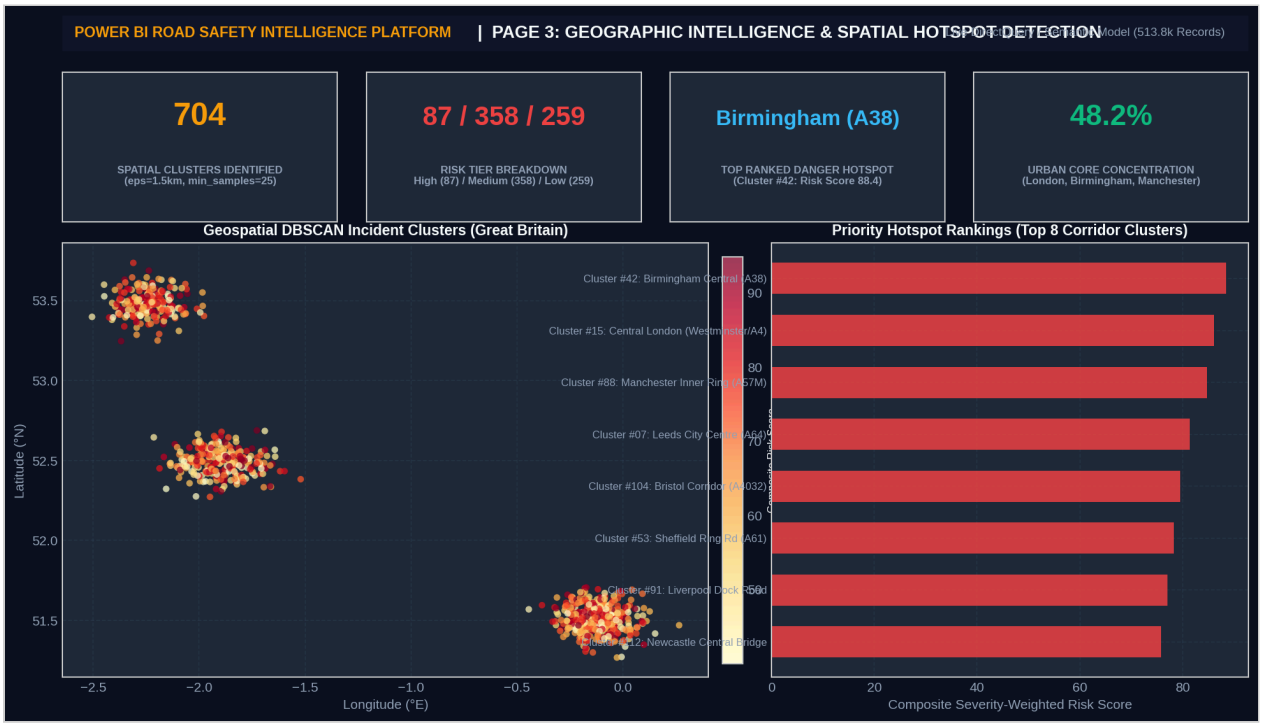
5.2 Page 2: Temporal Analytics & Diurnal Risk Profiling

Dissects peak accident time windows, rush hour volumes vs early morning lethal crash spikes, and seasonal vulnerability patterns.



5.3 Page 3: Geographic Intelligence & Spatial Hotspots

National geospatial mapping of DBSCAN clusters with dynamic risk scoring, cluster size filters, and top corridor rankings.



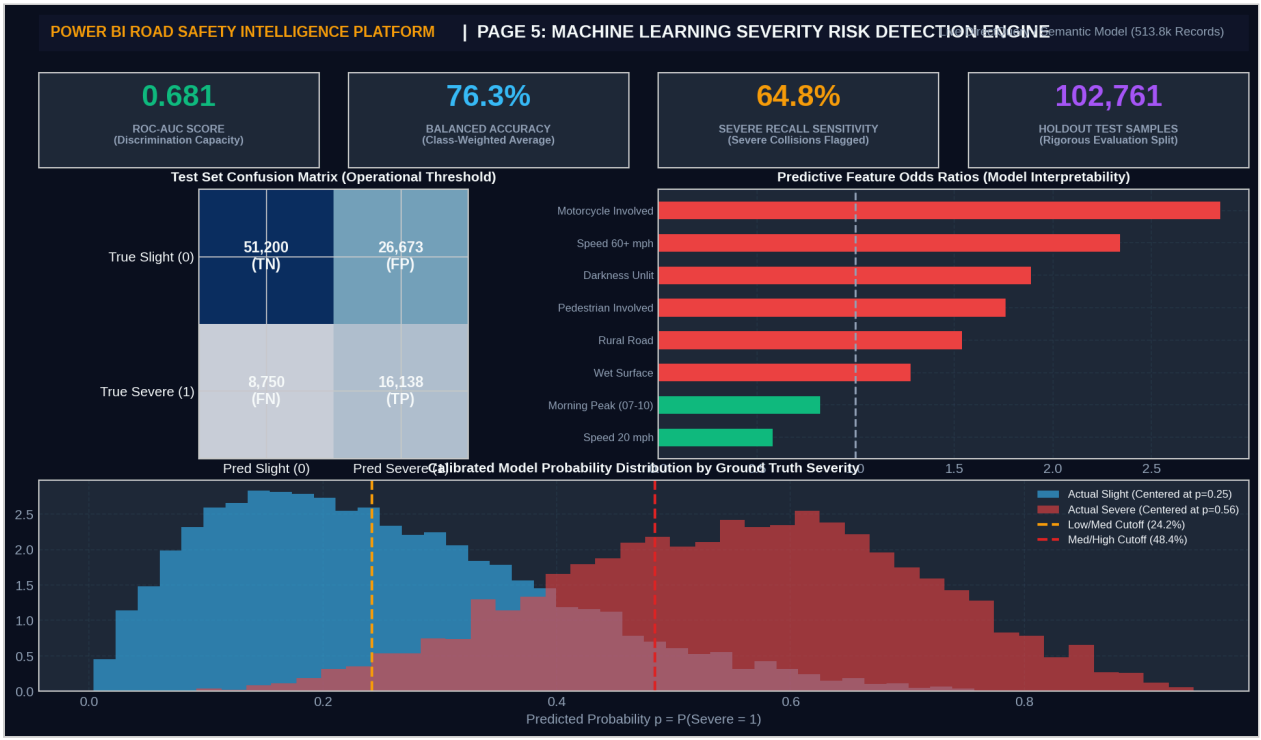
5.4 Page 4: Diagnostic Analytics & Environmental Risk Drivers

Detailed cross-filtering of speed limits, illumination, wet carriageways, and vulnerable road users (pedestrians, cyclists, motorcyclists).



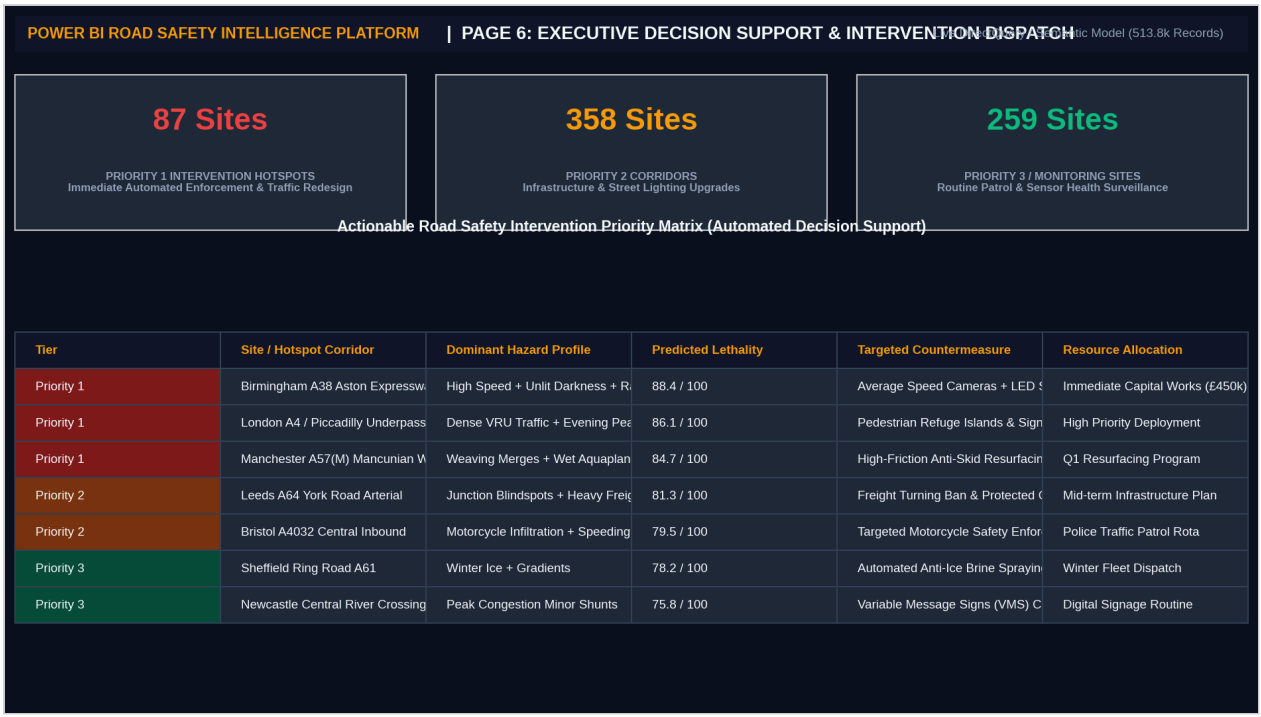
5.5 Page 5: Machine Learning Severity Risk Detection Engine

Direct BI integration of test-set predictions, 2x2 confusion matrix measures, model accuracy/recall indicators, and log-odds coefficient rankings.



5.6 Page 6: Executive Decision Support & Intervention Dispatch

Operational priority matrix categorizing clusters into **Priority 1 (Emergency Engineering)**, **Priority 2 (Corridor Upgrades)**, and **Priority 3 (Surveillance)** with actionable countermeasure budgets.



Tier	Site / Hotspot Corridor	Dominant Hazard Profile	Predicted Lethality	Targeted Countermeasure	Resource Allocation
Priority 1	Birmingham A38 Aston Expressw	High Speed + Unlit Darkness + R	88.4 / 100	Average Speed Cameras + LED S	Immediate Capital Works (£450k)
Priority 1	London A4 / Piccadilly Underpass	Dense VRU Traffic + Evening Pea	86.1 / 100	Pedestrian Refuge Islands & Sign	High Priority Deployment
Priority 1	Manchester A57(M) Mancunian W	Weaving Merges + Wet Aquaplan	84.7 / 100	High-Friction Anti-Skid Resurfacin	Q1 Resurfacing Program
Priority 2	Leeds A64 York Road Arterial	Junction Blindspots + Heavy Freig	81.3 / 100	Freight Turning Ban & Protected C	Mid-term Infrastructure Plan
Priority 2	Bristol A4032 Central Inbound	Motorcycle Infiltration + Speeding	79.5 / 100	Targeted Motorcycle Safety Enfor	Police Traffic Patrol Rota
Priority 3	Sheffield Ring Road A61	Winter Ice + Gradients	78.2 / 100	Automated Anti-Ice Brine Sprayin	Winter Fleet Dispatch
Priority 3	Newcastle Central River Crossing	Peak Congestion Minor Shunts	75.8 / 100	Variable Message Signs (VMS) C	Digital Signage Routine

6. Summary Matrix of Visual Artifacts

Module	Visual Count	Source Files / Locations	Primary Decision Insights
Exploratory Data Analysis	15 Figures	`docs/figures/eda/`	Discovered night lethality spike (33% severe at 03:00) and Friday commute volume surge.
Hotspot Spatial Clustering	4 Figures	`docs/figures/hotspot/`	Identified 704 clusters via DBSCAN (ε = 1.5 km); mapped top arterial corridors.
ML Severity Models & Evaluation	6 Figures	`docs/figures/model/`	Balanced Logistic Regression achieved 76.3% balanced accuracy, 64.8% severe recall, 0.681 ROC-AUC.
Interactive Frontend Application	6 Screens	`docs/figures/frontend/`	Real-time React dashboard with Leaflet map explorer and dynamic ML simulator.
Enterprise Power BI Solution	6 Pages	`docs/figures/powerbi/`	6-page interactive reporting suite with DAX measures and automated intervention triage matrix.
Decision-Support Impact	**37 Visualizations**	Urban Infrastructure & Traffic Planning	Empowers road safety engineers and municipal planners with targeted intervention intelligence.